Pacific-Basin Finance Journal 87 (2024) 102483

Contents lists available at ScienceDirect

Pacific-Basin Finance Journal

journal homepage: www.elsevier.com/locate/pacfin

An extension analysis of Amihud's illiquidity premium: Evidence from the Taiwan stock market

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ARTICLE INFO

Keywords:
Stock illiquidity ratio
Overnight returns
Recency effect
Asset pricing
Long–short strategy

ABSTRACT

This study proposes a novel illiquidity measure that excludes overnight returns and incorporates the recency effect, and investigates whether this new metric more effectively captures the illiquidity premium in the Taiwan stock market relative to Amihud's (2002) measure and other illiquidity measures that exclude overnight returns or account for the recency effect. Cross-sectional and time-series asset pricing tests reveal that our measure dominates other illiquidity measures, yielding a more profitable long–short strategy. Vector autoregressive analysis confirms these results for the aggregate market. Our study contributes to the measurement of liquidity in emerging markets and suggests potential improvements for portfolio management and asset pricing models.

1. Introduction

The objective of this paper is to examine the influence of illiquidity measures on the cross-section and time-series of expected stock returns in the Taiwan stock market. Although Amihud (2002) stock illiquidity measure has been widely used in recent studies, this measure suffers from some limitations. Specifically, Amihud's stock illiquidity ratio has a shortcoming because of a mismatch between the denominator and numerator, which can be attributed to overnight returns (Barardehi et al., 2021). Furthermore, Amihud's stock illiquidity ratio uses equal-weighted daily ratios to calculate the illiquidity ratio for a given period, and the recency effect is thus neglected in its computation (Greenwood and Shleifer, 2014; Barberis et al., 2018). The study proposes a novel approach to alleviate the limitations related to the denominator–numerator mismatch and address the recency effect when calculating illiquidity ratios. Specifically, the current paper employs four stock illiquidity measures to address the problem of the stock illiquidity premium in the Taiwan stock market. The four illiquidity measures comprise Amihud (2002) illiquidity ratio based on close-to-close returns, time-weighted Amihud (2002) illiquidity ratio that accounts for the recency effect, daytime Amihud (2002) illiquidity ratio based on open-to-close returns (excluding overnight returns), and time-weighted daytime Amihud (2002) illiquidity ratio which excludes overnight returns and accounts for the recency effect. To examine the challenges associated with the denominator–numerator mismatch and the recency effect, the present study compares these illiquidity measures with respect to their explanatory power regarding the extent of the illiquidity premium and their ability to subsume other illiquidity ratios.

Prior studies have intensively employed the illiquidity ratio suggested by Amihud (2002) to explore the effect of stock illiquidity on

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https://doi.org/10.1016/j.pacfin.2024.102483
Received 15 April 2024; Received in revised form 31 July 2024; Accepted 9 August 2024
Available online 10 August 2024
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Table 1
Descriptive statistics for close-to-close, overnight, and daytime returns.

	Close-to-Close	Overnight	Daytime
Mean	0.0191	0.0546	−0.0320
Min	−0.0406	−0.0223	−0.0503
Max	0.0933	0.1126	−0.0098
Std	0.0293	0.0280	0.0116

This table provides descriptive statistics for monthly close-to-close, overnight, and daytime returns. This study decomposes daily stock return into their overnight and daytime components, following the approach introduced by Lou et al. (2019). For each firm i on day d, we define the close-to-close return as follows:

$$R_{i,d}^{CTC} = \frac{P_{i,d}^{C}}{P_{i,d-1}^{C}} - 1$$

Where $P_{i,d}^{C}$ is the closing price on day d. The daytime (open-to-close) return, expressed as the relative price change from the market opening to closing on day d, is defined as:

$$R_{i,d}^{OTC} = \frac{P_{i,d}^{C}}{P_{i,d}^{O}} - 1$$

where $P_{i,d}^{C}$ is defined as described earlier and $P_{i,d}^{O}$ is the opening price. We then calculate the overnight (close-to-open) return from the daytime return ($R_{i,d}^{OTC}$) and daily close-to-close return ($R_{i,d}^{CTC}$) as:

$$R_{i,d}^{CTO} = \frac{1 + R_{i,d}^{CTC}}{1 + R_{i,d}^{OTC}} - 1$$

The opening and closing prices are adjusted by dividends, stock splits, and other corporate actions. The study then accumulates the close-to-close, daytime, and overnight returns across the days within month m:

$$R_{i,m}^{OTC} = \prod_{d \in m} \left(1 + R_{i,d,m}^{OTC}\right) - 1$$

$$R_{i,m}^{CTO} = \prod_{d \in m} \left(1 + R_{i,d,m}^{CTO}\right) - 1$$

$$R_{i,m}^{CTC} = \left(1 + R_{i,m}^{OTC}\right)\left(1 + R_{i,m}^{CTO}\right) - 1$$

where $R_{i,d,m}^{CTC}$, $R_{i,d,m}^{CTO}$, and $R_{i,d,m}^{OTC}$ are the close-to-close, overnight, and daytime returns for stock i on day d within a month m. $R_{i,m}^{CTC}$, $R_{i,m}^{CTO}$, and $R_{i,m}^{OTC}$ are the close-to-close, overnight, and daytime returns for stock i in month m. The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.

expected stock returns (Kamara et al., 2008; Watanabe and Watanabe, 2008; Chordia et al., 2009; Florackis et al., 2011; Liang and Wei, 2012; Brennan et al., 2013; Lou and Shu, 2017; Amihud and Noh, 2021a; Amihud and Noh, 2021b; Barardehi et al., 2021). Amihud's illiquidity measure has three distinct advantages over many other illiquidity measures. First, it is straightforward to use, enabling efficient computations for long time horizons because of the widespread availability of return and trading volume data. Additionally, its intuitive appeal stems from the fact that the average daily price response associated with a dollar of trading volume enables it to be an accurate proxy for Kyle's price impact coefficient. Finally, the positive return premium of Amihud's illiquidity measure is typically regarded as an illiquidity premium that compensates for price impact.

Although widely adopted as a straightforward price impact measure, Amihud (2002) ratio *has* some limitations. *One* key limitation concerns the mismatch between the denominator and numerator of Amihud's illiquidity ratio. Barardehi et al. (2021) highlight that the denominator reflects trading volume during trading hours only, whereas the numerator captures the absolute return between the previous day's close and the current day, including overnight price movements. This creates a fundamental time mismatch within the ratio. The inclusion of overnight returns in Amihud (2002) stock illiquidity measure renders it potentially unsuitable for the *Taiwan stock market*. Barardehi et al. (2021) suggest that overnight returns are associated with public information and are uncorrelated *with* trading costs. With the increasing integration of international equity markets, any news concerning U.S. stock markets may help foreign investors (e.g., particularly those in Asian markets) infer future price *movements* in their domestic stock markets based on current U.S. stock market price movements. Thus, the closing returns of the U.S. stock market may affect the overnight returns of Asian markets. Fung et al. (2010) find that the overnight returns of Asian markets are closely correlated with the closing returns of the U.S. stock market, suggesting *that* the overnight returns of Asian markets are closely related to public information rather than trading costs. The Taiwan stock market is mostly dominated by technology stocks, *which are* frequently influenced by the performance of the NASDAQ and Philadelphia Semiconductor indices. This influence is particularly noteworthy given that TSMC (Taiwan Semiconductor Manufacturing Company), Taiwan's largest company by market capitalization, is a constituent stock of the Philadelphia Semiconductor Index. The effects of the U.S. stock markets are more pronounced on the opening prices of Taiwan's markets than on those of other countries' stock markets. Consequently, the inclusion of overnight returns *in* the calculation of Amihud's stock illiquidity measure may lead to *severely biased estimates for the Taiwan stock market.*

Another problem with Amihud (2002) stock illiquidity measure relates to the use of equal-weighted daily ratios to calculate the illiquidity ratio over a given period (e.g., weekly or monthly). According to several influential studies, the recency effect shapes the beliefs and decisions of investors by leading them to pay more attention to recent events (Bhootra and Hur, 2013; Greenwood and

Table 2
Descriptive statistics for illiquidity measures.

	$ILLIQ^{A}_{i,m}$	$ILLIQ^{TA}_{i,m}$	$ILLIQ^{DA}_{i,m}$	$ILLIQ^{TDA}_{i,m}$
Mean	0.0183	0.0180	0.0166	0.0165
Min	0.0076	0.0075	0.0079	0.0079
Max	0.0558	0.0552	0.0367	0.0369
Std	0.0123	0.0120	0.0074	0.0074

This table reports descriptive statistics for Amihud's illiquidity ratio ($ILLIQ^{A}_{i,m}$; see Amihud, 2002), time-weighted Amihud's illiquidity ratio ($ILLIQ^{TA}_{i,m}$), the daytime Amihud's illiquidity ratio ($ILLIQ^{DA}_{i,m}$; see Barardehi et al., 2021), and time-weighted daytime Amihud's illiquidity ratio ($ILLIQ^{TDA}_{i,m}$). The two illiquidity measures for time-weighted adjustment mentioned above, which take into account the recency effect, are proposed in this paper. The illiquidity ratio proposed by Amihud (2002) is calculated as follows:

$$ILLIQ^{A}_{i,m} = \frac{1}{D_{i,m}} \sum_{d=1}^{D_{i,m}} \frac{\left|R^{CTC}_{i,d,m}\right|}{DVOL_{i,d,m}}$$

where $R^{CTC}_{i,d,m}$ is the close-to-close return for stock i on day d within a month m, $DVOL_{i,d,m}$ is the dollar trading volume for stock i on day d within a month m, and $D_{i,m}$ is the number of trading days, which the trading occurs over the month m, for stock i. Prior literature suggests that individuals tend to put more weight on recent information in comparison to distant information due to the recency effect (see Greenwood and Shleifer, 2014; Barberis et al., 2018). As such, this paper proposes the time-weighted Amihud's illiquidity ratio as follows:

$$ILLIQ^{TA}_{i,m} = \sum_{d=1}^{D_{i,m}} \frac{W_{i,d,m}}{\sum_{d=1}^{D_{i,m}} W_{i,d,m}} \times \frac{\left|R^{CTC}_{i,d,m}\right|}{DVOL_{i,d,m}}$$

where $W_{i,d,m}$ is the time-weighted function and is defined as (see Chen et al., 2022):

$$W_{i,d,m} = \frac{2}{1 + exp\left[\alpha - \frac{2\alpha d}{D_{i,m} \times \eta}\right]}$$

where $\alpha \geq 0$ is called the adjustment parameter or the discount rate. When $\alpha = 0$, the weight $W_{i,d,m} = 1$, which means that time has no influence on the loss function. When $\alpha > 0$, the closer to the current time, the larger the weight is, and the more obvious the influence of time on the loss function. This paper chooses $\alpha = 0.9$. To alleviate the impact of the overnight information on the Amihud (2002) illiquidity measure, Barardehi et al. (2021) propose the daytime (open-to-close) Amihud's illiquidity measure as follows:

$$ILLIQ^{DA}_{i,m} = \frac{1}{D_{i,m}} \sum_{d=1}^{D_{i,m}} \frac{\left|R^{OTC}_{i,d,m}\right|}{DVOL_{i,d,m}}$$

where $R^{OTC}_{i,d,m}$ is the open-to-close return for stock i on day d within each month m and $DVOL_{i,d,m}$ as well as $D_{i,m}$ are defined analogously to $ILLIQ^{A}_{i,m}$. Again, this paper considers the daytime Amihud's illiquidity measure adjusted for the recency effect, named the time-weighted daytime (open-to-close) Amihud's illiquidity measure, as follows:

$$ILLIQ^{TDA}_{i,m} = \sum_{d=1}^{D_{i,m}} \frac{W_{i,d,m}}{\sum_{d=1}^{D_{i,m}} W_{i,d,m}} \times \frac{\left|R^{OTC}_{i,d,m}\right|}{DVOL_{i,d,m}}$$

The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.

Shleifer, 2014; Barberis et al., 2018; Gandre, 2020; Mohrschladt, 2021).[1,2] This effect also influences the stock price behavior in the Taiwan stock market. An influential study by Hao et al. (2016) indicates that the recency strategy does indeed exist in the Taiwan stock market. Hao et al. (2016) reveal that the recency strategy still gives rise to predictable time-series patterns when conditioning variables are included, with the strategy being profitable during expansionary periods and up-market states. Similarly, Wang and Lien (2023) report that recency effect affects asset pricing in the Taiwan stock market. Wang and Lien (2023) indicate that the strength of the lottery anomaly diminishes when the 52-week high occurs more recently, implying that investors place more weight on recent information regarding the 52-week high. The findings by Hao et al. (2016) and Wang and Lien (2023) suggest that the recency effect plays a significant role in the price behavior of the Taiwan stock market. As such, the present study suggests that the recency effect should be considered when calculating the illiquidity ratio in the Taiwan stock market.

Building upon previous research, the present study introduces a novel measure, the time-weighted daytime Amihud's illiquidity ratio, which excludes overnight returns and incorporates the recency effect. The primary objective is to assess whether this innovative measure more effectively captures the illiquidity premium in the Taiwan stock market compared with the traditional Amihud's

[1] Early experimental evidence on the effect of information ordering is provided by Murdock (1962), whose focus is on psychology. In that study, participants were presented with a list of words in a free-recall experiment where the order of words was irrelevant. The participants nonetheless exhibited a greater tendency of recalling words at the end of the list.

[2] According to the recency effect, recent observations have a stronger impact on the formation of beliefs compared to observations from the more distant past. Therefore, traders rely less on stock-specific reference points from distant past experiences due to this effect. Consequently, the recency effect shapes the decision-making process in financial markets. Bansal et al. (2023) suggest that traders attach declining weight to reference points from the distant past, possibly due to the recency effect.

Table 3
Portfolio returns sorting based on illiquidity measures.

	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	$P_{10}-P_1$
Panel A: Equal-weighted portfolio returns											
$ILLIQ^{A}_{i,m}$	0.0042	0.0062	0.0057	0.0065	0.0083	0.0096	0.0091	0.0119	0.0173	0.0291	0.0249***
$ILLIQ^{TA}_{i,m}$	0.0041	0.0059	0.0061	0.0067	0.0078	0.0095	0.0093	0.0119	0.0173	0.0293	0.0252***
$ILLIQ^{DA}_{i,m}$	0.0046	0.0065	0.0061	0.0067	0.0076	0.0094	0.0091	0.0105	0.0171	0.0304	0.0259***
$ILLIQ^{TDA}_{i,m}$	0.0045	0.0067	0.0062	0.0067	0.0072	0.0097	0.0088	0.0105	0.0165	0.0312	0.0267***
Panel B: Value-weighted portfolio returns											
$ILLIQ^{A}_{i,m}$	0.0044	0.0062	0.0059	0.0067	0.0084	0.0096	0.0092	0.0118	0.0171	0.0284	0.0240***
$ILLIQ^{TA}_{i,m}$	0.0044	0.0059	0.0062	0.0069	0.0080	0.0096	0.0094	0.0118	0.0170	0.0285	0.0242***
$ILLIQ^{DA}_{i,m}$	0.0048	0.0064	0.0062	0.0068	0.0077	0.0095	0.0091	0.0104	0.0170	0.0296	0.0249***
$ILLIQ^{TDA}_{i,m}$	0.0047	0.0066	0.0063	0.0068	0.0073	0.0099	0.0089	0.0105	0.0163	0.0304	0.0257***

This table presents portfolio returns based on Amihud's illiquidity ratio ($ILLIQ^{A}_{i,m}$; see Amihud, 2002), time-weighted Amihud's illiquidity ratio ($ILLIQ^{TA}_{i,m}$), the daytime Amihud's illiquidity ratio ($ILLIQ^{DA}_{i,m}$; see Barardehi et al., 2021), and time-weighted daytime Amihud's illiquidity ratio ($ILLIQ^{TDA}_{i,m}$). The two illiquidity measures for time-weighted adjustment mentioned above, which take into account the recency effect (see Greenwood and Shleifer, 2014; Barberis et al., 2018), are proposed in this paper. Stocks are grouped into ten portfolios according to these illiquidity measures. This paper sorts stocks into decile portfolios according to their illiquidity measures and computes both equal-weighted (EW) and value-weighted (VW) portfolio returns for the subsequent month, denoted as $m+1$. Additionally, it provides monthly return for the zero-cost strategy, which involves buying high-illiquidity stocks (decile 10, P_{10}) and shorting low-illiquidity stocks (decile 1, P_1). Panel A presents the equal-weighted portfolio returns. Panel B presents the value-weighted portfolio returns. The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

Table 4
Risk-adjusted portfolio returns of the zero-cost strategy sorting based on illiquidity measures.

	Three-Factor Model	Four-Factor Model	Five-Factor Model	Six-Factor Model
Panel A: Equal-weighted approach				
$ILLIQ^{A}_{i,m}$	0.0196***	0.0226***	0.0210***	0.0235***
$ILLIQ^{TA}_{i,m}$	0.0198***	0.0228***	0.0211***	0.0237***
$ILLIQ^{DA}_{i,m}$	0.0205***	0.0231***	0.0216***	0.0238***
$ILLIQ^{TDA}_{i,m}$	0.0215***	0.0240***	0.0229***	0.0250***
Panel B: Value-weighted approach				
$ILLIQ^{A}_{i,m}$	0.0184***	0.0214***	0.0198***	0.0224***
$ILLIQ^{TA}_{i,m}$	0.0186***	0.0216***	0.0199***	0.0225***
$ILLIQ^{DA}_{i,m}$	0.0193***	0.0219***	0.0204***	0.0226***
$ILLIQ^{TDA}_{i,m}$	0.0203***	0.0228***	0.0217***	0.0238***

This table displays the risk-adjusted portfolio returns of the zero-cost strategy, utilizing Amihud's illiquidity ratio ($ILLIQ^{A}_{i,m}$; see Amihud, 2002), time-weighted Amihud's illiquidity ratio ($ILLIQ^{TA}_{i,m}$), the daytime Amihud's illiquidity ratio ($ILLIQ^{DA}_{i,m}$; see Barardehi et al., 2021), and time-weighted daytime Amihud's illiquidity ratio ($ILLIQ^{TDA}_{i,m}$). The two illiquidity measures for time-weighted adjustment mentioned above, which take into account the recency effect (see Greenwood and Shleifer, 2014; Barberis et al., 2018), are proposed in this paper. The Risk-adjusted portfolio returns of the zero-cost strategy based on illiquidity measures is calculated as follows. First, stocks are grouped into ten portfolios according to these illiquidity measures. This paper sorts stocks into decile portfolios according to their illiquidity measures and computes both equal-weighted (EW) and value-weighted (VW) portfolio returns for the subsequent month, denoted as $m+1$. Then, this paper provides monthly return for the zero-cost strategy, which involves buying high-illiquidity stocks (decile 10, P_{10}) and shorting low-illiquidity stocks (decile 1, P_1). Finally, the return on the zero-cost strategy is calculated by adjusting risk factors based on the three-factor model (Fama and French, 1993), the four-factor model (Carhart, 1997), the five-factor model (Fama and French, 2015), and the six-factor model (Fama and French, 2018). Panels A and B present the equal-weighted and value-weighted approaches, respectively. The t-statistics are based on Newey-West robust standard errors. The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

Table 5
Correlations for illiquidity measures and firm characteristics.

	$R^{CTC}_{i,m+1}$	$ILLIQ^{A}_{i,m}$	$ILLIQ^{TA}_{i,m}$	$ILLIQ^{DA}_{i,m}$	$ILLIQ^{TDA}_{i,m}$	$SIZE_{i,m}$	$BM_{i,m}$	$MOM_{i,m}$
$R^{CTC}_{i,m+1}$	1.0000	0.0512	0.0527	0.0674	0.0688	−0.0028	0.0253	0.0090
$ILLIQ^{A}_{i,m}$	0.0512	1.0000	0.9968	0.8651	0.8615	−0.0705	0.2283	−0.0326
$ILLIQ^{TA}_{i,m}$	0.0527	0.9968	1.0000	0.8690	0.8698	−0.0728	0.2381	−0.0384
$ILLIQ^{DA}_{i,m}$	0.0674	0.8651	0.8690	1.0000	0.9966	−0.0803	0.2594	−0.0425
$ILLIQ^{TDA}_{i,m}$	0.0688	0.8615	0.8698	0.9966	1.0000	−0.0823	0.2685	−0.0488
$SIZE_{i,m}$	−0.0028	−0.0705	−0.0728	−0.0803	−0.0823	1.0000	−0.0969	−0.0033
$BM_{i,m}$	0.0253	0.2283	0.2381	0.2594	0.2685	−0.0969	1.0000	−0.2588
$MOM_{i,m}$	0.0090	−0.0326	−0.0384	−0.0425	−0.0488	−0.0033	−0.2588	1.0000

This table reports the correlation matrix among the stock returns, four illiquidity measures, and other variables of firm characteristics. $R^{CTC}_{i,m+1}$ is the close-to-close return for stock i in month $m+1$. The four illiquidity measures are $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$. $ILLIQ^{D}_{i,m}$ is the Amihud's illiquidity ratio for stock i in month m. $ILLIQ^{TD}_{i,m}$ is the time-weighted Amihud's illiquidity ratio for stock i in month m. $ILLIQ^{DA}_{i,m}$ is the daytime Amihud's illiquidity ratio for stock i in month m. $ILLIQ^{TDA}_{i,m}$ is the time-weighted daytime Amihud's illiquidity for stock i in month m. $SIZE_{i,m}$ is the market capitalization for stock i at the end of month m. $BM_{i,m}$ is the book-to-market value for stock i at the end of month m. $MOM_{i,m}$ is the compound returns over past six-month at the end of m. The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.

Table 6
Fama-Macbeth cross-sectional regressions.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)	Model (7)	Model (8)	Model (9)
$ILLIQ^{A}_{i,m}$	0.4549***				-5.5038***	−0.2591	−0.3426		
$ILLIQ^{TA}_{i,m}$		0.4777***			6.0057***			−0.2326	
$ILLIQ^{DA}_{i,m}$			0.6528***			0.8623***			-2.1657***
$ILLIQ^{TDA}_{i,m}$				0.6786***			0.9656***	0.8600***	2.8519***
$SIZE_{i,m}$	−0.2026*	−0.1886*	−0.1345	-0.1176	−0.1152	−0.1163	−0.0992	−0.1010	−0.0876
$BM_{i,m}$	0.1842**	0.1764**	0.1443*	0.1361	0.1454*	0.1487*	0.1408	0.1403	0.1228
$MOM_{i,m}$	0.1822*	0.1839*	0.1853*	0.1888*	0.1990**	0.1922*	0.1984**	0.1957**	0.1996**
$AdjR$	0.0448	0.0449	0.0446	0.0447	0.0486	0.0491	0.0494	0.0492	0.0474

This table presents the Fama-Macbeth (see Fama and MacBeth, 1973) mean monthly coefficients for the impact of four illiquidity measures on subsequent stock close-to-close returns with controlling *SIZE*, *BM*, and *MOM*. The dependent variable, $R^{CTC}_{i,m+1}$, is the subsequent stock close-to-close return for firm i in month $m+1$. The independent variables are the $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, $ILLIQ^{TDA}_{i,m}$, $SIZE_{i,m}$, $BM_{i,m}$, and $MOM_{i,m}$. $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ are the four illiquidity measures, which represent the Amihud's illiquidity ratio, the time-weighted Amihud's illiquidity ratio, the daytime Amihud's illiquidity ratio, and the time-weighted daytime Amihud's illiquidity, respectively, for stock i in month m as considered in this paper. $BM_{i,m}$ is the book-to-market value for stock i at the end of month m. $MOM_{i,m}$ is the compound returns over past six-month at the end of m. In these regressions, all independent variables are converted into z-scores separately in each cross section. Each variable is mean zero with unit standard deviation. The intercept for each specification is not shown above, for brevity. The t-statistics are calculated based on Newey-West robust standard errors of the mean monthly coefficients (Newey and West, 1987). The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

illiquidity ratio. To disentangle the effects of excluding overnight returns and incorporating the recency effect, two additional analyses are conducted. First, a measure focusing solely on the recency effect (time-weighted Amihud's illiquidity measure) is examined. Second, a measure that only excludes overnight returns (daytime Amihud's illiquidity measure) is analyzed. These measures are compared against the original Amihud's illiquidity measure in terms of their effectiveness in explaining the illiquidity premium in the Taiwan stock market. Finally, the present study investigates whether the proposed time-weighted daytime Amihud's illiquidity ratio exhibits superior explanatory power for the illiquidity premium relative to both the time-weighted and daytime Amihud's illiquidity measures. Through this comprehensive analysis, the aim is to validate the robustness and efficacy of the novel measure. Following Florackis et al. (2011), the study employs a cross-sectional analysis and a time-series risk-adjusted approach to conduct asset pricing tests. These tests allow for the examination of whether the time-weighted daytime Amihud's illiquidity ratio subsumes the profitability of other illiquidity ratios.

Our cross-sectional and time-series asset pricing test results indicate that the time-weighted daytime Amihud's illiquidity ratio has a more pronounced effect on expected stock returns relative to other illiquidity measures. This finding suggests that the profitability associated with the time-weighted daytime Amihud's illiquidity measure surpasses that of other illiquidity measures, highlighting its stronger association with the illiquidity premium. The dominance of the time-weighted daytime Amihud's illiquidity measure is not

Table 7
Examining profitability of long-short strategies among illiquidity measures.

	K = *Equal-weighted*				K = *Value-weighted*			
	Three-Factor	Four-Factor	Five-Factor	Six-Factor	Three-Factor	Four-Factor	Five-Factor	Six-Factor
Panel A: Amihud's illiquidity ratio vs. time-weighted Amihud's illiquidity ratio								
$\zeta = A$ and $\xi = TA$	−0.0003	−0.0003	−0.0003	−0.0003	−0.0002	−0.0003	−0.0002	−0.0002
$\zeta = TA$ and $\xi = A$	0.0007**	0.0008**	0.0007**	0.0009**	0.0006**	0.0008**	0.0006*	0.0008**
Panel B: Amihud's illiquidity ratio vs. daytime Amihud's illiquidity ratio								
$\zeta = A$ and $\xi = DA$	−0.0006	0.0004	0.0000	0.0010	−0.0006	0.0004	0.0000	0.0011
$\zeta = DA$ and $\xi = A$	0.0021**	0.0015**	0.0016**	0.0010	0.0021**	0.0016**	0.0017**	0.0011
Panel C: Amihud's illiquidity ratio vs. time-weighted daytime Amihud's illiquidity ratio								
$\zeta = A$ and $\xi = TDA$	−0.0013	−0.0002	−0.0010	0.0002	−0.0014*	−0.0002	−0.0010	0.0003
$\zeta = TDA$ and $\xi = A$	0.0032***	0.0026***	0.0031***	0.0024***	0.0033***	0.0027***	0.0032***	0.0025***
Panel D: time-weighted Amihud's illiquidity ratio vs. time-weighted daytime Amihud's illiquidity ratio								
$\zeta = TA$ and $\xi = TDA$	−0.0010	0.0002	−0.0006	0.0006	−0.0011	0.0002	−0.0007	0.0006
$\zeta = TDA$ and $\xi = TA$	0.0026***	0.0018***	0.0025***	0.0016**	0.0028***	0.0020***	0.0027***	0.0018**
Panel E: Daytime Amihud's illiquidity ratio vs. time-weighted daytime Amihud's illiquidity ratio								
$\zeta = DA$ and $\xi = TDA$	−0.0007**	−0.0006*	−0.0010***	−0.0008**	−0.0007**	−0.0006*	-0.0010***	−0.0008**
$\zeta = TDA$ and $\xi = DA$	0.0011***	0.0011***	0.0015***	0.0014***	0.0012***	0.0011***	0.0015***	0.0014***

This table reports the intercepts of time-series regressions that examine the profitability of long-short strategies among illiquidity measures. Panel A reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between Amihud's illiquidity ratio and time-weighted Amihud's illiquidity ratio. Panel B reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between Amihud's illiquidity ratio and daytime Amihud's illiquidity ratio. Panel C reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between Amihud's illiquidity ratio and time-weighted daytime Amihud's illiquidity ratio. Panel D reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between time-weighted Amihud's illiquidity ratio and time-weighted daytime Amihud's illiquidity ratio. Panel E reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between daytime Amihud's illiquidity ratio and time-weighted daytime Amihud's illiquidity ratio. Specifically, for each Panel, this study employs the regression model suggested by Chen et al. (2021) as follows

$$R^{\zeta}_{LS,K,m} = \alpha_{\zeta} + \beta_1 R^{\xi}_{LS,K,m} + \beta_2 R^{MKT}_m + \beta_3 R^{SMB}_m + \beta_4 R^{HML}_m + \beta_5 R^{RMW}_m + \beta_6 R^{CAM}_m + \epsilon^{\zeta}_{LS,k,m}, K = VW \text{ or } EW$$

where $R^{\zeta}_{LS,K,m}$ represents the long-short portfolio return based on $ILLIQ^{\zeta}_{i,m}$ illiquidity ratio with a K-weighting approach in month m. $R^{\xi}_{LS,K,m}$ is the long-short portfolio return based on $ILLIQ^{\xi}_{i,m}$ illiquidity ratio with a K-weighting approach in month m. K is the equal-weighted or value-weighted approach. $\epsilon^{\zeta}_{LS,K,m}$ is the residual for the regression model. ζ and ξ represent different liquidity measures for stock i in month m. ζ and ξ are the elements of the set $\{A, TA, DA, TDA\}$, and it is required that $\zeta \neq \xi$. R^{MKT}_m, R^{SMB}_m, and R^{HML}_m are the market excess return, return of the small-minus-big (SMB) factor, and return of the high-minus-low (HML) factor, respectively, in month m. These factors are the Fama and French (1993) three-factor model. R^{RMW}_m and R^{CAM}_m the operating profitability and investment factors that are two additional factors in the Fama and French (2015) five-factor model in month m. R^{MOM}_m is the momentum factor, proposed by Carhart (1997), in month m. The p-values are calculated from the t-statistics which are corrected based on Newey-West robust standard errors (Newey and West, 1987). The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

only due to the recency effect but also to the exclusion of overnight returns. Additionally, the present study employs vector autoregressive (VAR) analysis to distinguish causality and predictability between Amihud's illiquidity ratio and the time-weighted daytime Amihud's illiquidity measure over short- and long- time horizons. The empirical evidence reveals that both Amihud's illiquidity and the time-weighted daytime Amihud's illiquidity can Granger-cause aggregate market returns, but not vice versa. Notably, the time-weighted daytime Amihud's illiquidity measure exhibits a more pronounced positive effect on subsequent aggregate market returns relative to Amihud's illiquidity measure.

This paper makes several important contributions to the literature on liquidity measurement and asset pricing. First, to the best of our knowledge, our study is the first to comprehensively analyze the components of the illiquidity premium by separately examining the effects of overnight return exclusion and time weighting.[3] By investigating the effect of overnight returns on illiquidity ratio computation in the Taiwan stock market, the present study deepens the understanding of how these returns affect the accurate measurement of price impact and trading costs. This contributes to the literature on the illiquidity premium in non-U.S. markets,

[3] The authors are grateful to the reviewer for this insightful suggestion, which has enhanced the depth of this research.

Table 8
VAR results for the market return and the measures of illiquidity ratios.

	$y = R_m^{MKT}$		$y = AILLIQ_m^A$		$y = AILLIQ_m^{TDA}$	
Panel A: Causality tests						
	$\delta_{1l}(L) = 0$	$\gamma_{1l}(L) = 0$	$\beta_{2l}(L) = 0$	$\gamma_{2l}(L) = 0$	$\beta_{3l}(L) = 0$	$\delta_{3l}(L) = 0$
With 1 lag	5.4527**	0.1025	1.2059	14.1053***	0.7438	1.6154
With 3 lags	8.4300***	8.8697***	0.9219	6.1173***	1.6037	2.6780**
With 6 lags	5.4666***	5.1404***	0.6491	4.8946***	1.3509	3.5815**
With 9 lags	6.0560***	4.2868***	0.2074	1.5474	0.9860	5.7025***
With 12 lags	6.9744***	4.2946***	0.1400	0.7403	1.0658	11.2947***
Panel B: Sum of coefficients tests						
	$\sum_{l=1}^{L}\delta_{1l}$	$\sum_{l=1}^{L}\gamma_{1l}$	$\sum_{l=1}^{L}\beta_{2l}$	$\sum_{l=1}^{L}\gamma_{2l}$	$\sum_{l=1}^{L}\beta_{3l}$	$\sum_{l=1}^{L}\delta_{3l}$
With 1 lag	−0.0051**	0.2441	−0.2376	1.9457***	−0.0136	0.0017
With 3 lags	0.0057	0.7968	−0.1965	2.3518***	−0.0557	−0.0004
With 6 lags	−0.0014	0.9794**	−0.3130	2.3629***	-0.0788*	−0.0012
With 9 lags	−0.0001	1.4030***	−0.6538	1.9826***	-0.0500*	−0.0010
With 12 lags	0.0363*	2.3540***	0.3781	3.5476	-0.0187	-0.0032**

This table presents the VAR results for the market return and the measures of illiquidity ratios. Panel A reports the Granger-causality tests and Panel B displays the sum of coefficients tests. The VAR is estimated as follows:

$$R_m^{MKT} = \alpha_1 + \sum_{l=1}^{L}\beta_{1l}R_{m-l}^{MKT} + \sum_{l=1}^{L}\delta_{1l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{1l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{MKT}$$

$$AILLIQ_m^{A} = \alpha_2 + \sum_{l=1}^{L}\beta_{2l}R_{m-l}^{MKT} + \sum_{l=1}^{L}\delta_{2l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{2l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^A}$$

$$AILLIQ_m^{TDA} = \alpha_3 + \sum_{l=1}^{L}\beta_{3l}R_{m-l}^{MKT} + \sum_{l=1}^{L}\delta_{3l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{3l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^{TDA}}$$

where R_m^{MKT} is the market return in month m. $AILLIQ_m^A$ is the average of Amihud's illiquidity ratio across stocks in each month and is calculated as:

$$AILLIQ_m^{A} = \frac{1}{N_m}\sum_{i=1}^{N_m} ILLIQ_{i,m}^{A}$$

where N_m is the number of stocks in month m. $ILLIQ_{i,m}^A$ is the Amihud's illiquidity ratio for stock i in moth m. $AILLIQ_m^{TDA}$ is the average of the time-weighted daytime Amihud's illiquidity ratio in each month and is calculated in an analogous manner to $AILLIQ_m^A$ as:

$$AILLIQ_m^{TDA} = \frac{1}{N_m}\sum_{i=1}^{N_m} ILLIQ_{i,m}^{TDA}$$

where N_m is the number of stocks in month m and $ILLIQ_{i,m}^{TDA}$ is the time-weighted daytime Amihud's illiquidity ratio for stock i in moth m. ϵ_m^{MKT}, $\epsilon_m^{ILLIQ^A}$, and $\epsilon_m^{ILLIQ^{TDA}}$ are the residuals for R_m^{MKT}, $AILLIQ_m^A$, and $AILLIQ_m^{TDA}$. The Newey–West standard errors are used to compute F-statistics. The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

particularly Asian markets, where overnight returns are closely correlated with the closing returns of the U.S. stock market (e.g., Fung et al., 2010). Furthermore, the examination of whether the illiquidity premium can be enhanced by considering time weighting contributes to recent research on recency effect–based investment strategies and asset pricing (e.g., Greenwood and Shleifer, 2014; Hao et al., 2016; Barberis et al., 2018; Gandre, 2020; Mohrschladt, 2021). Second and most important, based on the above findings, this paper introduces the time-weighted daytime Amihud's illiquidity ratio, a novel liquidity measure that addresses two key limitations of the traditional Amihud's measure by excluding overnight returns and incorporating the recency effect in the Taiwan stock market. This innovative measure can more accurately represent intraday illiquidity dynamics and the illiquidity risk premium compared to other illiquidity measures examined in this study. The comparative analysis of the proposed measure against traditional and partially modified Amihud's measures not only validates the robustness of the new indicator but also challenges existing methodologies in liquidity measurement, potentially providing insights for portfolio management, risk assessment, and asset pricing models in emerging markets.

The remainder of the present article is organized as follows. Section 2 describes the research data and the construction of the illiquidity measures. Section 3 presents the empirical results. Section 4 concludes with the key findings and implications of the present study.

2. Data, decomposition of stock returns, and measures of stock illiquidity

2.1. Data

Our sample comprises all common stocks listed on the Taiwan Stock Exchange (TWSE) and Taipei Exchange (TPE) from January

Table 9
VAR results for the futures return and the measures of illiquidity ratios.

	$y = R_m^{FUT}$		$y = AILLIQ_m^A$		$y = AILLIQ_m^{TDA}$	
Panel A: Causality tests						
	$\delta_{1l}(L) = 0$	$\gamma_{1l}(L) = 0$	$\beta_{2l}(L) = 0$	$\gamma_{2l}(L) = 0$	$\beta_{3l}(L) = 0$	$\delta_{3l}(L) = 0$
With 1 lag	7.3426***	0.0918	1.0558	10.5497***	0.7830	1.6208
With 3 lags	13.8642***	7.0875***	0.7756	7.3192***	1.8222	2.7451**
With 6 lags	6.4672***	3.6272***	0.6118	2.9306***	1.1308	2.7353**
With 9 lags	4.3774***	4.1610***	0.2123	1.1536	0.9750	5.4563***
With 12 lags	4.3148***	3.5223***	0.1329	0.4948	0.9100	7.1062**
Panel B: Sum of coefficients tests						
	$\sum_{l=1}^{L}\delta_{1l}$	$\sum_{l=1}^{L}\gamma_{1l}$	$\sum_{l=1}^{L}\beta_{2l}$	$\sum_{l=1}^{L}\gamma_{2l}$	$\sum_{l=1}^{L}\beta_{3l}$	$\sum_{l=1}^{L}\delta_{3l}$
With 1 lag	−0.0063***	0.2232	−0.3542	1.7819***	−0.0127	0.0017
With 3 lags	0.0093	0.7271	−0.3590	2.3009***	−0.0590	−0.0005
With 6 lags	0.0073	0.7981*	−0.4814	2.4991***	-0.0912*	−0.0010
With 9 lags	0.0079	1.2703***	−0.5195	2.1100**	-0.0491*	−0.0008
With 12 lags	0.0416**	2.2410***	0.6945	3.5970	-0.0082	-0.0031**

This table presents the VAR results for the futures and the measures of illiquidity ratios. Panel A reports the Granger-causality tests and Panel B displays the sum of coefficients tests. The VAR is estimated as follows:

$$R_m^{FUT} = \alpha_1 + \sum_{l=1}^{L}\beta_{1l}R_{m-l}^{FUT} + \sum_{l=1}^{L}\delta_{1l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{1l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{FUT}$$

$$AILLIQ_m^A = \alpha_2 + \sum_{l=1}^{L}\beta_{2l}R_{m-l}^{FUT} + \sum_{l=1}^{L}\delta_{2l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{2l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^A}$$

$$AILLIQ_m^{TDA} = \alpha_3 + \sum_{l=1}^{L}\beta_{3l}R_{m-l}^{FUT} + \sum_{l=1}^{L}\delta_{3l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{3l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^{TDA}}$$

where R_m^{FUT} is the futures return in month m. This paper utilizes the futures index associated with the Taiwan Weighted Stock Index traded on the Taiwan Futures Exchange (TAIFEX) for analysis. $AILLIQ_m^A$ is the average of Amihud's illiquidity ratio across stocks in each month and is calculated as:

$$AILLIQ_m^A = \frac{1}{N_m}\sum_{i=1}^{N_m} ILLIQ_{i,m}^A$$

where N_m is the number of stocks in month m. $ILLIQ_{i,m}^A$ is the Amihud's illiquidity ratio for stock i in moth m. $AILLIQ_m^{TDA}$ is the average of the time-weighted daytime Amihud's illiquidity ratio in each month and is calculated in an analogous manner to $AILLIQ_m^A$ as:

$$AILLIQ_m^{TDA} = \frac{1}{N_m}\sum_{i=1}^{N_m} ILLIQ_{i,m}^{TDA}$$

where N_m is the number of stocks in month m and $ILLIQ_{i,m}^{TDA}$ is the time-weighted daytime Amihud's illiquidity ratio for stock i in moth m. ϵ_m^{FUT}, $\epsilon_m^{ILLIQ^A}$, and $\epsilon_m^{ILLIQ^{TDA}}$ are the residuals for R_m^{FUT}, $AILLIQ_m^A$, and $AILLIQ_m^{TDA}$. The Newey–West standard errors are used to compute F-statistics. The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

2008 to October 2023.[4] Daily data for opening and closing prices, trading volume in dollars, and firm characteristics are collected from the *Taiwan Economic Journal* (*TEJ*), which is a local data vendor in Taiwan. All prices, returns, trading volume in dollars, and financial variables are calculated in the local currency (the New Taiwan dollar). The paper uses the Taiwan Central Bank discount rate as the proxy for the risk-free rate.

2.2. Decomposition of stock returns

We decompose daily stock return into overnight and daytime components in accordance with the methodology of Lou et al. (2019). Specifically, for each firm i on day d, we define the close-to-close return as follows:

$$R_{i,d}^{CTC} = \frac{P_{i,d}^C}{P_{i,d-1}^C} - 1 \tag{1}$$

where $P_{i,d}^C$ is the closing price on day d. The daytime (open-to-close) return, expressed as the relative price change from the market opening to closing on day d, is defined as

$$R_{i,d}^{OTC} = \frac{P_{i,d}^C}{P_{i,d}^O} - 1 \tag{2}$$

[4] The reason for selecting a sample from 2008 is the availability of the TEJ data retrieval API (Application Programming Interface) starting in that year. The TEJ data retrieval API enables practitioners to effortlessly access data, develop trading strategies, construct asset pricing models, and identify return anomalies.

Table 10
Robustness check: Fama-Macbeth cross-sectional regressions.

	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)	Model (7)	Model (8)	Model (9)
$ILLIQ^{A}_{i,m}$	0.4549***				-4.8162***	−0.2591	−0.3506		
$ILLIQ^{TA}_{i,m}$		0.4806***			5.3195***			−0.2249	
$ILLIQ^{DA}_{i,m}$			0.6528***			0.8623***			-1.8892***
$ILLIQ^{TDA}_{i,m}$				0.6815***			0.9758***	0.8552***	2.5769***
$SIZE_{i,m}$	−0.2026*	-0.1869*	−0.1345	−0.1157	−0.1148	−0.1163	−0.0975	−0.0993	−0.0871
$BM_{i,m}$	0.1842**	0.1754*	0.1443*	0.1351	0.1454*	0.1487*	0.1399	0.1394	0.1224
$MOM_{i,m}$	0.1822*	0.1841*	0.1853*	0.1893*	0.1992**	0.1922*	0.1992**	0.1961**	0.1997**
$AdjR$	0.0448	0.0449	0.0446	0.0447	0.0486	0.0491	0.0494	0.0492	0.0474

This table presents the robustness check for the Fama-Macbeth (see Fama and French (1993)) mean monthly coefficients for the impact of four illiquidity measures on subsequent stock returns with controlling *SIZE*, *BM*, and *MOM*. The dependent variable, $R^{CTC}_{i,m+1}$, is the subsequent stock close-to-close return for firm i in month $m+1$. The independent variables are the $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, $ILLIQ^{TDA}_{i,m}$, $SIZE_{i,m}$, $BM_{i,m}$, and $MOM_{i,m}$. $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ are the four illiquidity measures, which represent the Amihud's illiquidity ratio, the time-weighted Amihud's illiquidity ratio, the daytime Amihud's illiquidity ratio, and the time-weighted daytime Amihud's illiquidity, respectively, for stock i in month m as considered in this paper. For our robustness check, we adhere to the approach suggested by Chen et al. (2022), where we set $\alpha = 1$ when computing the time-weighted function for Amihud's illiquidity and daytime Amihud's illiquidity ratios. $BM_{i,m}$ is the book-to-market value for stock i at the end of month m. $MOM_{i,m}$ is the compound returns over past six-month at the end of m. In these regressions, all independent variables are converted into z-scores separately in each cross section. Each variable is mean zero with unit standard deviation. The intercept for each specification is not shown above, for brevity. The t-statistics are calculated based on Newey-West robust standard errors of the mean monthly coefficients (Newey and West, 1987). The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

where $P^{C}_{i,d}$ is defined as described earlier and $P^{O}_{i,d}$ is the opening price. We then calculate the overnight (close-to-open) return from the daytime return ($R^{OTC}_{i,d}$) and daily close-to-close return ($R^{CTC}_{i,d}$) by applying the following equations:

$$R^{CTO}_{i,d} = \frac{1 + R^{CTC}_{i,d}}{1 + R^{OTC}_{i,d}} - 1 \tag{3}$$

Opening and closing prices are adjusted by dividends, stock splits, and other corporate actions. The study then accumulates the close-to-close, daytime, and overnight returns across the days within month m by applying the following equations:

$$R^{OTC}_{i,m} = \prod_{d \in m} \left(1 + R^{OTC}_{i,d,m}\right) - 1 \tag{4}$$

$$R^{CTO}_{i,m} = \prod_{d \in m} \left(1 + R^{CTO}_{i,d,m}\right) - 1 \tag{5}$$

$$R^{CTC}_{i,m} = \left(1 + R^{OTC}_{i,m}\right)\left(1 + R^{CTO}_{i,m}\right) - 1 \tag{6}$$

where $R^{CTC}_{i,d,m}$, $R^{CTO}_{i,d,m}$, and $R^{OTC}_{i,d,m}$ are the close-to-close, overnight, and daytime returns for stock i on day d within month m. $R^{CTC}_{i,m}$, $R^{CTO}_{i,m}$, and $R^{OTC}_{i,m}$ are the close-to-close, overnight, and daytime returns, respectively, for stock i in month m.

2.3. Measures of stock illiquidity

The illiquidity ratio proposed by Amihud (2002) is calculated as follows:

$$ILLIQ^{A}_{i,m} = \frac{1}{D_{i,m}} \sum_{d=1}^{D_{i,m}} \frac{\left|R^{CTC}_{i,d,m}\right|}{DVOL_{i,d,m}} \tag{7}$$

where $R^{CTC}_{i,d,m}$ is the close-to-close return for stock i on day d within month m, $DVOL_{i,d,m}$ is the dollar trading volume for stock i on day d within month m, and $D_{i,m}$ is the number of trading days for stock i in month m. As indicated in Eq. (7), Amihud (2002) stock illiquidity measure calculates the average of the daily absolute close-to-close return per dollar traded over a given period spanning D consecutive trading days, where daily close-to-close returns incorporate public overnight information.

According to Barardehi et al. (2021), Amihud's illiquidity ratio introduces a crucial time misalignment between its numerator and denominator, undermining the reliability of the measure for capturing market liquidity. Specifically, the denominator measures the

Table 11
Robustness check: Examining profitability of long-short strategies among illiquidity measures.

	K = *Equal-weighted*				K = *Value-weighted*			
	Three-Factor	Four-Factor	Five-Factor	Six-Factor	Three-Factor	Four-Factor	Five-Factor	Six-Factor
Panel A: Amihud's illiquidity ratio vs. time-weighted Amihud's illiquidity ratio								
$\zeta = A$ and $\xi = TA$	−0.0006*	−0.0006	−0.0006*	−0.0006	−0.0005*	−0.0005	−0.0005	−0.0005
$\zeta = TA$ and $\xi = A$	0.0011***	0.0012***	0.0012***	0.0013***	0.0010***	0.0011***	0.0011***	0.0012***
Panel B: Amihud's illiquidity ratio vs. daytime Amihud's illiquidity ratio								
$\zeta = A$ and $\xi = DA$	−0.0006	0.0004	0.0000	0.0010	−0.0006	0.0004	0.0000	0.0011
$\zeta = DA$ and $\xi = A$	0.0021***	0.0015**	0.0016**	0.0010	0.0021***	0.0016**	0.0017**	0.0011
Panel C: Amihud's illiquidity ratio vs. time-weighted daytime Amihud's illiquidity ratio								
$\zeta = A$ and $\xi = TDA$	−0.0015*	−0.0003	−0.0010	0.0001	−0.0015*	−0.0003	−0.0010	0.0002
$\zeta = TDA$ and $\xi = A$	0.0033***	0.0027***	0.0031***	0.0024***	0.0033***	0.0027***	0.0032***	0.0025***
Panel D: time-weighted Amihud's illiquidity ratio vs. time-weighted daytime Amihud's illiquidity ratio								
$\zeta = TA$ and $\xi = TDA$	−0.0007	0.0004	−0.0002	0.0009	−0.0008	0.0004	−0.0003	0.0009
$\zeta = TDA$ and $\xi = TA$	0.0024***	0.0017***	0.0021***	0.0014*	0.0025***	0.0018***	0.0023***	0.0015*
Panel E: Daytime Amihud's illiquidity ratio vs. time-weighted daytime Amihud's illiquidity ratio								
$\zeta = DA$ and $\xi = TDA$	−0.0008***	−0.0007**	-0.0011***	-0.0009**	−0.0008***	-0.0007**	-0.0010***	-0.0008**
$\zeta = TDA$ and $\xi = DA$	0.0013***	0.0012***	0.0015***	0.0015***	0.0013***	0.0012***	0.0015***	0.0014***

This table reports the intercepts of time-series regressions that examine the profitability of long-short strategies among illiquidity measures. As suggested Chen et al. (2022), $\alpha = 1$ is adopted for our robustness check as calculating time-weighted function for Amihud's illiquidity ratio. Panel A reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between Amihud's illiquidity ratio and time-weighted Amihud's illiquidity ratio. Panel B reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between Amihud's illiquidity ratio and daytime Amihud's illiquidity ratio. Panel C reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between Amihud's illiquidity ratio and time-weighted daytime Amihud's illiquidity ratio. Panel D reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between time-weighted Amihud's illiquidity ratio and time-weighted daytime Amihud's illiquidity ratio. Panel E reports the results of the time-series regressions for the value- and equal-weighted long-short portfolio returns between daytime Amihud's illiquidity ratio and time-weighted daytime Amihud's illiquidity ratio. Specifically, for each Panel, this study employs the regression model suggested by Chen et al. (2021) as follows

$$R^{\zeta}_{LS,K,m} = \alpha_{\zeta} + \beta_1 R^{\xi}_{LS,K,m} + \beta_2 R^{MKT}_m + \beta_3 R^{SMB}_m + \beta_4 R^{HML}_m + \beta_5 R^{RMW}_m + \beta_6 R^{CAM}_m + \epsilon^{\zeta}_{LS,k,m}, K = VW \text{ or } EW.$$

where $R^{\zeta}_{LS,K,m}$ represents the long-short portfolio return based on $ILLIQ^{\zeta}_{i,m}$ illiquidity ratio with a K-weighting approach in month m. $R^{\xi}_{LS,K,m}$ is the long-short portfolio return based on $ILLIQ^{\xi}_{i,m}$ illiquidity ratio with a K-weighting approach in month m. K is the equal-weighted or value-weighted approach. $\epsilon^{\zeta}_{LS,K,m}$ is the residual for the regression model. ζ and ξ represent different liquidity measures for stock i in month m. ζ and ξ are the elements of the set $\{A, TA, DA, TDA\}$, and it is required that $\zeta \neq \xi$. R^{MKT}_m, R^{SMB}_m, and R^{HML}_m are the market excess return, return of the small-minus-big (SMB) factor, and return of the high-minus-low (HML) factor, respectively, in month m. These factors are the Fama and French (1993) three-factor model. R^{RMW}_m and R^{CAM}_m the operating profitability and investment factors that are two additional factors in the Fama and French (2015) five-factor model in month m. R^{MOM}_m is the momentum factor, proposed by Carhart (1997), in month m. The p-values are calculated from the t-statistics which are corrected based on Newey-West robust standard errors (Newey and West, 1987). The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

dollar trading volume during regular trading hours, whereas the numerator reflects the absolute return from the close of the previous day to the close of the current day, including overnight price movements. However, overnight returns are influenced by public information and do not correlate with trading costs, introducing a significant time misalignment in Amihud's illiquidity ratio (Barardehi et al., 2021). To address the problem of overnight information affecting Amihud (2002) illiquidity measure, Barardehi et al. (2021) propose a modification known as the daytime (open-to-close) Amihud's illiquidity measure, which is defined as follows:

$$ILLIQ^{DA}_{i,m} = \frac{1}{D_{i,m}} \sum_{d=1}^{D_{i,m}} \frac{\left| R^{OTC}_{i,d,m} \right|}{DVOL_{i,d,m}} \tag{8}$$

where $R^{OTC}_{i,d,m}$ is the open-to-close return for stock i on day d within each month m. $DVOL_{i,d,m}$ as well as $D_{i,m}$ are defined analogously to $ILLIQ^{CTC}_{i,m}$.

The recency effect plays a key role in financial markets (Greenwood and Shleifer, 2014; Barberis et al., 2018), including the Taiwan stock market (Hao et al., 2016; Wang and Lien, 2023). Accordingly, the present study introduces a time-weighted approach and

Table 12
Robustness check: VAR results for the futures return and the measures of illiquidity ratios.

	$y = R_m^{FUT}$		$y = AILLIQ_m^A$		$y = AILLIQ_m^{TDA}$	
Panel A: Causality tests						
	$\delta_{1l}(L) = 0$	$\gamma_{1l}(L) = 0$	$\beta_{2l}(L) = 0$	$\gamma_{2l}(L) = 0$	$\beta_{3l}(L) = 0$	$\delta_{3l}(L) = 0$
With 1 lag	7.4047***	0.0904	1.0461	10.5088***	0.6721	1.5927
With 6 lags	6.4982***	3.7337***	0.6133	2.9523***	1.0766	2.6932**
With 12 lags	4.3229***	3.5578***	0.1343	0.4989	0.8987	42.2950***
Panel B: Sum of coefficients tests						
	$\sum_{l=1}^{L}\delta_{1l}$	$\sum_{l=1}^{L}\gamma_{1l}$	$\sum_{l=1}^{L}\beta_{2l}$	$\sum_{l=1}^{L}\gamma_{2l}$	$\sum_{l=1}^{L}\beta_{3l}$	$\sum_{l=1}^{L}\delta_{3l}$
With 1 lag	−0.0063***	0.2205	−0.3527	1.7815***	−0.0118	0.0017
With 6 lags	0.0074	0.7995*	−0.4945	2.4949***	-0.0921*	−0.0011
With 12 lags	0.0416**	2.2314***	0.6827	3.5628	-0.0077	-0.0031**

This table presents the robustness check of VAR results for the futures and the measures of illiquidity ratios. For our robustness check, we adhere to the approach suggested by Chen et al. (2022), where we set $\alpha = 1$ when computing the time-weighted function for Amihud's illiquidity ratio. Panel A reports the Granger-causality tests and Panel B displays the sum of coefficients tests. The VAR is estimated as follows:

$$R_m^{FUT} = \alpha_1 + \sum_{l=1}^{L}\beta_{1l}R_{m-l}^{FUT} + \sum_{l=1}^{L}\delta_{1l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{1l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{FUT}$$

$$AILLIQ_m^A = \alpha_2 + \sum_{l=1}^{L}\beta_{2l}R_{m-l}^{FUT} + \sum_{l=1}^{L}\delta_{2l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{2l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^A}$$

$$AILLIQ_m^{TDA} = \alpha_3 + \sum_{l=1}^{L}\beta_{3l}R_{m-l}^{FUT} + \sum_{l=1}^{L}\delta_{3l}AILLIQ_{m-l}^{A} + \sum_{l=1}^{L}\gamma_{3l}AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^{TDA}}$$

where R_m^{FUT} is the futures return in month m. This paper utilizes the futures index associated with the Taiwan Weighted Stock Index traded on the Taiwan Futures Exchange (TAIFEX) for analysis. $AILLIQ_m^A$ is the average of Amihud's illiquidity ratio across stocks in each month and is calculated as:

$$AILLIQ_m^A = \frac{1}{N_m}\sum_{i=1}^{N_m} ILLIQ_{i,m}^A$$

where N_m is the number of stocks in month m. $ILLIQ_{i,m}^A$ is the Amihud's illiquidity ratio for stock i in moth m. $AILLIQ_m^{TDA}$ is the average of the time-weighted daytime Amihud's illiquidity ratio in each month and is calculated in an analogous manner to $AILLIQ_m^A$ as:

$$AILLIQ_m^{TDA} = \frac{1}{N_m}\sum_{i=1}^{N_m} ILLIQ_{i,m}^{TDA}$$

where N_m is the number of stocks in month m and $ILLIQ_{i,m}^{TDA}$ is the time-weighted daytime Amihud's illiquidity ratio for stock i in moth m. ϵ_m^{FUT}, $\epsilon_m^{ILLIQ^A}$, and $\epsilon_m^{ILLIQ^{TDA}}$ are the residuals for R_m^{FUT}, $AILLIQ_m^A$, and $AILLIQ_m^{TDA}$. The Newey–West standard errors are used to compute F-statistics. The sample contains common stocks listed on the TWSE (the Taiwan Stock Exchange) and Taipei Exchange (TPE) from January 2008 to October 2023.
*, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

provides the revised versions of Eqs. (7) and (8) as follows:

$$ILLIQ_{i,m}^{TA} = \sum_{d=1}^{D_{i,m}} \frac{W_{i,d,m}}{\sum_{d=1}^{D_{i,m}} W_{i,d,m}} \times \frac{\left|R_{i,d,m}^{CTC}\right|}{DVOL_{i,d,m}} \tag{9}$$

$$ILLIQ_{i,m}^{TDA} = \sum_{d=1}^{D_{i,m}} \frac{W_{i,d,m}}{\sum_{d=1}^{D_{i,m}} W_{i,d,m}} \times \frac{\left|R_{i,d,m}^{OTC}\right|}{DVOL_{i,d,m}} \tag{10}$$

where $ILLIQ_{i,m}^{TA}$ is the time-weighted Amihud's illiquidity measure, $ILLIQ_{i,m}^{TDA}$ is the time-weighted daytime (open-to-close) Amihud's illiquidity measure, and $W_{i,d,m}$ is the time-weighted function and is defined as follows (Chen et al., 2022):

$$W_{i,d,m} = \frac{2}{1 + exp\left[\alpha - \frac{2\alpha d}{D_{i,m}\times\eta}\right]} \tag{11}$$

where $\alpha \geq 0$ is the adjustment parameter or the discount rate. If $\alpha = 0$, the weight $W_{i,d,m} = 1$, indicating that time does not influence the loss function. When $\alpha > 0$, the closer to the current time, the larger the weight is, and the more obvious the influence of time on the loss function. This study uses $\alpha = 0.9$.

3. Empirical results

3.1. Preliminary statistics

3.1.1. Preliminary stock returns and illiquidity

Table 1 presents the descriptive statistics for monthly close-to-close, overnight, and daytime returns spanning from January 2008 to October 2023. The means of the monthly average close-to-close and overnight returns are 0.0191 and 0.0546, respectively, whereas the mean of the monthly average daytime return is −0.0320. These results indicate that the positive monthly average close-to-close returns are primarily driven by the positive monthly average overnight returns. Table 1 also indicates that the absolute value of the monthly average overnight returns is higher than that of the monthly average daytime returns.[5] Furthermore, the standard deviations of close-to-close, overnight, and daytime returns are 0.0293, 0.0280, and 0.0116, respectively. Overall, Table 1 suggests that the monthly average close-to-close returns are significantly influenced by overnight returns, with the latter being primarily attributed by public information. This effect is particularly pronounced in the Taiwan stock market, where overnight returns are mainly influenced by the stock returns of the U.S. stock markets on the previous day. Barardehi et al. (2021) highlight that overnight returns and transaction costs are not directly related. Therefore, a necessary step is excluding overnight returns when calculating the illiquidity ratio.

Table 2 presents the descriptive statistics for four illiquidity metrics: namely Amihud's illiquidity ratio (Amihud, 2002; $ILLIQ^{A}_{i,m}$,), the time-weighted Amihud's illiquidity ratio ($ILLIQ^{TA}_{i,m}$), the daytime Amihud's illiquidity ratio (Barardehi et al., 2021; $ILLIQ^{DA}_{i,m}$), and the time-weighted daytime Amihud's illiquidity ratio ($ILLIQ^{TDA}_{i,m}$). The two time-weighted illiquidity measures that account for the recency effect proposed in the present study are $ILLIQ^{TA}_{i,m}$and $ILLIQ^{TDA}_{i,m}$ (Greenwood and Shleifer, 2014; Barberis et al., 2018). Our analysis reveals that the value of $ILLIQ^{A}_{i,m}$ is higher than those of $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$. Specifically, the means of $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$ and $ILLIQ^{TDA}_{i,m}$ are 0.0183, 0.0180, 0.0166, and 0.0165, respectively. Furthermore, Table 2 reveals that the standard deviation of $ILLIQ^{A}_{i,m}$ is greater than those of the other three measures. Similar results are obtained for the means and standard deviations of $ILLIQ^{DA}_{i,m}$ and $ILLIQ^{TDA}_{i,m}$. Overall, the results in Table 2 align with the findings of Barardehi et al. (2021), suggesting that incorporating overnight price movements tends to exaggerate the illiquidity measure for most stocks.

3.1.2. Portfolio approach to illiquidity ratios

Table 3 reports returns of portfolios constructed from $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$. This paper sorts stocks into decile portfolios according to their illiquidity measures and computes both equal-weighted (EW) and value-weighted (VW) portfolio returns for the subsequent month, denoted as $m+1$.[6] We also provide monthly return for the zero-cost strategy, which involves buying high-illiquidity stocks (decile 10, P_{10}) and shorting low-illiquidity stocks (decile 1, P_1).

Panel A presents the equal-weighted portfolio returns. When the portfolio is constructed using $ILLIQ^{A}_{i,m}$, the returns for portfolio 10 (P_{10}) and portfolio 1 (P_1) are 0.0291 and 0.0042, respectively. The difference between the two portfolio returns, $P_{10}-P_1$, is 0.0249, which is significantly positive at the 1% level. For the portfolio constructed with $ILLIQ^{TA}_{i,m}$, portfolio 10 (P_{10}) yields a return of 0.0293, whereas portfolio 1 (P_1) yields a return of 0.0041. The difference in return between portfolio 10 and portfolio 1, $P_{10}-P_1$, is 0.0252, which is statistically significant at the 1% level. Similarly, the differences in the values of $P_{10}-P_1$ for $ILLIQ^{DA}_{i,m}$ and $ILLIQ^{TDA}_{i,m}$ portfolios are both significantly positive at 0.0259 and 0.0267, respectively. The results align with those of prior studies (e.g., Amihud, 2002; Barardehi et al., 2021; among others), indicating that the illiquidity premium prevails in the Taiwan stock market. Furthermore, the long–short profitability exhibits a sequential increase from $ILLIQ^{A}_{i,m}$ to $ILLIQ^{TA}_{i,m}$ portfolios, then from $ILLIQ^{TA}_{i,m}$ to $ILLIQ^{DA}_{i,m}$ portfolios, and finally from $ILLIQ^{DA}_{i,m}$ to $ILLIQ^{TDA}_{i,m}$ portfolios. Panel B presents the value-weighted portfolio returns. The results are similar to those obtained through the equal-weighted portfolios. Thus, regardless of whether an equal-weighted or value-weighted approach is used to calculate portfolio returns, the long–short strategy (buying high-illiquidity stocks and selling low-illiquidity stocks) is profitable on the basis of any of the four illiquidity measures, namely $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$. The orders of long–short portfolio returns, from the smallest to the largest, are those from $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ portfolios.

Table 4 presents the risk-adjusted portfolio returns through the four zero-cost strategies mentioned above. The return on the zero-

[5] A key study by French and Roll (1986) suggests that hourly daytime volatility is substantially higher than hourly overnight volatility because daytime trading induces volatility by revealing price-sensitive private information. However, recent evidence indicates that there is considerable public information accumulating overnight (see Tsiakas, 2008). This phenomenon is pronounced in the Taiwan stock market, where overnight returns incorporate public information from the preceding days in the US stock markets.

[6] Given the extreme skewness in market equity values, strict capitalization weighting can lead to distorted investment weights in certain months. This approach not only reflects an unrealistic investment policy but can also result in less precise point estimates due to the noise introduced by these less diversified portfolios. The extreme skewness in market capitalization in Taiwan exacerbates this issue. For example, TSMC alone constitutes approximately 35% of the total market capitalization of listed stocks, while the top 10 companies account for 50.53% of the overall market value. Therefore, this study adopts the approach of Loughran and Ritter (2000), Ikenberry and Ramnath (2002), and Boehme et al. (2009) by utilizing log value-weighted portfolios to mitigate the distortions caused by extreme capitalization weights. The market capitalization for top 10 companies can be referred to https://www.taifex.com.tw/cht/9/futuresQADetail.

cost strategy is calculated by adjusting risk factors based on the three-factor model (Fama and French, 1993), the four-factor model (Carhart, 1997), the five-factor model (Fama and French, 2015), and the six-factor model (Fama and French, 2018). Panels A and B illustrate the results from the equal-weighted and value-weighted approaches, respectively. As revealed in Table 4, both approaches yield significantly positive risk-adjusted alpha returns. The long–short strategy portfolios yield significantly positive returns, with the risk-adjusted alpha increasing from $ILLIQ^{A}_{i,m}$ to $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and ultimately $ILLIQ^{TDA}_{i,m}$. For example, when the value-weighted approach is employed with the six-factor model, the risk-adjusted alpha returns are significant and positive for $ILLIQ^{A}_{i,m}$ (0.0224), $ILLIQ^{TA}_{i,m}$ (0.0225), $ILLIQ^{DA}_{i,m}$ (0.0226), and $ILLIQ^{TDA}_{i,m}$ (0.0238). $ILLIQ^{TDA}_{i,m}$ produces the largest illiquidity premium, whereas $ILLIQ^{A}_{i,m}$ yields the smallest illiquidity premium. These findings are consistent with those in Table 3. Overall, the results in Tables 3 and 4 suggest that excluding the overnight return and incorporating the recency effect can enhance the returns of investment strategies when calculating illiquidity premiums in the Taiwan stock market.

3.2. Asset pricing tests

3.2.1. Cross-sectional regression analysis

A key benefit of the aforementioned portfolio analysis is that it does not rest on any assumptions about the specific functional form of the relationships between illiquidity measures and subsequent stock returns. However, aggregating stocks into portfolios leads to a loss of information because it masks differences across firms in terms of characteristics beyond those used for sorting. In this subsection, the paper employs firm-level Fama and MacBeth (1973) regressions to control for a broad range of firm characteristics simultaneously.

To investigate how effectively excluding the overnight return effect and incorporating the recency effect improve the capture of the illiquidity premium, this study sequentially evaluates (1) whether the time-weighted Amihud's illiquidity measure ($ILLIQ^{TA}_{i,m}$) outperforms the original Amihud's illiquidity measure ($ILLIQ^{A}_{i,m}$) in capturing the illiquidity premium, (2) whether the daytime Amihud's illiquidity measure ($ILLIQ^{DA}_{i,m}$) outperforms $ILLIQ^{A}_{i,m}$ in capturing the illiquidity premium, and (3) whether the time-weighted daytime Amihud's illiquidity measure ($ILLIQ^{TDA}_{i,m}$) subsumes the performance of $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, and $ILLIQ^{DA}_{i,m}$ in capturing the illiquidity premium. This paper estimates the predictive cross-sectional regression of stock returns in month $m+1$ on a firm's measures of illiquidity ratios, controlling for a vector of variables $V_{i,m}$ as measured at the end of month m:

$$R^{CTC}_{i,m+1} = \alpha_m + \beta_{1,m} ILLIQ^{\zeta}_{i,m} + \beta_{2,m} ILLIQ^{\xi}_{i,m} + \Pi'_m V_{i,m} + \mu_{i,m} \tag{12}$$

where ζ and ξ represent two distinct liquidity measures for stock i in month m. ζ and ξ are the elements of the set $\{A, TA, DA, TDA\}$ under the condition of $\zeta \neq \xi$. $R^{CTC}_{i,m+1}$, $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ are defined as previously described. $V_{i,m}$ includes $SIZE_{i,m}$, $BM_{i,m}$, and $MOM_{i,m}$ in the general specification. $SIZE_{i,m}$ is the market capitalization for stock i at the end of month m. $BM_{i,m}$ is the book-to-market value for stock i at the end of month m. $MOM_{i,m}$ is the compound returns over the past six-month at the end of m. In these regressions, all independent variables are converted into z-scores separately in each cross section. Each variable has a zero mean and a unit standard deviation. We estimate Eq. (12) for each month and subsequently calculate the time-series averages of the coefficients for the independent variables and the corresponding t-statistics by using the Newey–West robust standard errors of the mean monthly coefficients.

Before conducting the regression analysis, the study examines the correlations among variables to check for potential multicollinearity. The results are presented in Table 5. Except for the correlations among the illiquidity measures, $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$, all other correlations are low. For example, the correlation between $ILLIQ^{A}_{i,m}$ and $SIZE_{i,m}$is -0.0705. However, the correlation between $ILLIQ^{A}_{i,m}$ and $ILLIQ^{TDA}_{i,m}$ is high (0.8615). Therefore, a necessary step is identifying which variable among $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ is the most appropriate for representing the illiquidity factor.

Table 6 displays the Fama–Macbeth average monthly coefficients of $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ for subsequent cross-sectional stock returns, controlling *SIZE*, *BM*, and *MOM*. Models (1) through (4) analyze the separate effects of $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ on subsequent cross-sectional stock returns, accounting for the effects of *SIZE*, *BM*, and *MOM*. The results from Models (1) to (4) indicate that all four illiquidity measures significantly and positively influence subsequent cross-sectional stock returns, suggesting that an illiquidity premium exists in the Taiwan stock market. Notably, the regression coefficients for $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$ increase progressively from $ILLIQ^{A}_{i,m}$ to $ILLIQ^{TDA}_{i,m}$. Specifically, the regression coefficients are significantly positive for $ILLIQ^{A}_{i,m}$ (0.4549), $ILLIQ^{TA}_{i,m}$ (0.4777), $ILLIQ^{DA}_{i,m}$ (0.6528), and $ILLIQ^{TDA}_{i,m}$ (0.6786).

Model (5) introduces both $ILLIQ^{A}_{i,m}$ and $ILLIQ^{TA}_{i,m}$ into the regression framework to investigate whether the time-weighted Amihud's illiquidity measure ($ILLIQ^{TA}_{i,m}$) provides a more accurate representation of the illiquidity premium in stock returns relative to the original Amihud's illiquidity measure ($ILLIQ^{A}_{i,m}$). The results reveal that the regression coefficients for both $ILLIQ^{A}_{i,m}$ and $ILLIQ^{TA}_{i,m}$ are significant at the 1% level. When both $ILLIQ^{A}_{i,m}$ and $ILLIQ^{TA}_{i,m}$ are included in the regression model, the coefficient for $ILLIQ^{A}_{i,m}$ becomes negative, which is contrary to theoretical expectations, whereas the coefficient for $ILLIQ^{TA}_{i,m}$ remains positive. This result indicates that the time-weighted Amihud's illiquidity measure ($ILLIQ^{TA}_{i,m}$) is more effective than the original Amihud's illiquidity measure ($ILLIQ^{A}_{i,m}$) in capturing

the illiquidity premium in the stock market. Similarly, Model (6) includes both $ILLIQ^{A}_{i,m}$ and $ILLIQ^{DA}_{i,m}$ to test whether $ILLIQ^{DA}_{i,m}$ outperforms $ILLIQ^{A}_{i,m}$ in capturing the illiquidity premium. The empirical results reveal that the coefficient for $ILLIQ^{A}_{i,m}$ is not statistically significant, while the coefficient for $ILLIQ^{DA}_{i,m}$ is significantly positive. This result suggests that $ILLIQ^{DA}_{i,m}$ is more effective than $ILLIQ^{A}_{i,m}$ at capturing the illiquidity premium in the Taiwan stock market.

The final objective of the Fama–MacBeth regression model is to determine whether the liquidity measures $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, and $ILLIQ^{DA}_{i,m}$ are significantly different compared with $ILLIQ^{TDA}_{i,m}$, as specified in Models (7) through (9). This comparison is made to evaluate the magnitude of the illiquidity premium for each liquidity measure relative to $ILLIQ^{TDA}_{i,m}$. The empirical results from Models (7) to (9) indicate that the regression coefficients for the liquidity measures $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, and $ILLIQ^{DA}_{i,m}$ are all negative. Specifically, the coefficient for $ILLIQ^{DA}_{i,m}$ in Model (9) is significantly negative (−2.1657). By contrast, the regression coefficients for the liquidity measure $ILLIQ^{TDA}_{i,m}$ in Models (7) to (9) are significantly positive, with values of 0.9656, 0.8600, and 2.8519, respectively.

In summary, Table 6 presents empirical results indicating that the liquidity measures $ILLIQ^{TA}_{i,m}$ and $ILLIQ^{DA}_{i,m}$ outperform $ILLIQ^{A}_{i,m}$ in capturing the illiquidity premium. Furthermore, the study provides evidence that $ILLIQ^{TDA}_{i,m}$, which considers time weighting and the recency effect, provides a more comprehensive and effective approach for capturing the illiquidity premium compared with $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, and $ILLIQ^{DA}_{i,m}$. The empirical evidence suggests that the exclusion of the overnight return effect combined with the incorporation of the recency effect enables the effective measurement of the illiquidity premium in the Taiwan stock market.

3.2.2. Time-series regression analysis with factor models

Although the results obtained using Eq. (12) indicate that $ILLIQ^{TDA}_{i,m}$ can provide more relevant information relative to the other illiquidity measures for subsequent cross-section of stock returns, it provides limited insight into the long–short strategies based on $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, $ILLIQ^{DA}_{i,m}$, and $ILLIQ^{TDA}_{i,m}$. Examining the performance of long-short strategies offers investors an alternative approach to analyze and determine which illiquidity proxy variable yields more profitability and provides practical insights into portfolio formation.

In accordance with the approach of Fama and MacBeth (1973) for cross-sectional regressions, the present study investigates the relative performance of long–short strategies based on different illiquidity measures. Specifically, this study sequentially examines (1) whether the profitability of $ILLIQ^{TA}_{i,m}$ surpasses that of $ILLIQ^{A}_{i,m}$, (2) whether the profitability of $ILLIQ^{DA}_{i,m}$ encompasses that of $ILLIQ^{A}_{i,m}$, and (3) whether the profitability of $ILLIQ^{TDA}_{i,m}$ subsumes the performance of the other illiquidity measures (i.e., $ILLIQ^{A}_{i,m}$, $ILLIQ^{TA}_{i,m}$, and $ILLIQ^{DA}_{i,m}$). To empirically test the aforementioned three hypotheses outlined, the present study employs the regression model suggested by Chen et al. (2021), which are as follows:

$$R^{\zeta}_{LS,K,m} = \alpha_{\zeta} + \beta_1 R^{\xi}_{LS,K,m} + \beta_2 R^{MKT}_{m} + \beta_3 R^{SMB}_{m} + \beta_4 R^{HML}_{m} + \beta_5 R^{RMW}_{m} + \beta_6 R^{CAM}_{m} + \epsilon^{\zeta}_{LS,k,m}, K = VW \text{ or } EW \tag{13}$$

where $R^{\zeta}_{LS,K,m}$ represents the long–short portfolio return in month m based on the $ILLIQ^{\zeta}_{i,m}$ illiquidity ratio and K-weighting approach. $R^{\xi}_{LS,K,m}$ is the long–short portfolio return in month m based on the $ILLIQ^{\xi}_{i,m}$ illiquidity ratio and K-weighting approach. K is the equal-weighted or value-weighted approach. $\epsilon^{\zeta}_{LS,K,m}$ is the residual for the regression model. ζ and ξ represent two distinct liquidity measures for stock i in month m. ζ and ξ are the elements of the set $\{A, TA, DA, TDA\}$ under the condition $\zeta \neq \xi$. R^{MKT}_{m}, R^{SMB}_{m}, and R^{HML}_{m} are the market excess return, return of the small-minus-big (SMB) factor, and return of the high-minus-low (HML) factor, respectively, in month m. These factors are the Fama and French (1993) three-factor model. R^{RMW}_{m} and R^{CAM}_{m} the operating profitability and investment factors that are two additional factors in the Fama and French (2015) five-factor model in month m. R^{MOM}_{m} is the momentum factor, proposed by Carhart (1997), in month m. As shown in Eq. (13), a significantly positive intercept, α_{ζ}, suggests that the $ILLIQ^{\xi}_{i,m}$-based strategy does not subsume the profitability of the $ILLIQ^{\zeta}_{i,m}$-based strategy. Conversely, if α_{ζ} is not significantly positive, it suggests that the $ILLIQ^{\zeta}_{i,m}$-based strategy is subsumed by the $ILLIQ^{\xi}_{i,m}$-based strategy.

Table 7 reports the coefficients for the intercept α_{ζ} of time-series regressions that examine the profitability of long–short strategies across various illiquidity measures. Panel A of Table 7 presents the results of time-series regressions for the equal- and value-weighted returns of long–short portfolios, comparing Amihud's illiquidity ratio with the time-weighted Amihud's illiquidity ratio. For the results in the first row of Panel A in Table 7, we set $\zeta = A$ and $\xi = TA$ to verify whether the returns of long–short portfolios constructed using variable $ILLIQ^{A}_{i,m}$ can be dominated by portfolios constructed using $ILLIQ^{TA}_{i,m}$. The empirical results in the first row of Panel A in Table 7 indicate that neither the equal-weighted nor the value-weighted regression coefficients of α_A are not significantly different from zero. For the results in the second row of Panel A in Table 7, this study sets $\zeta = TA$ and $\xi = A$ to verify whether the illiquidity premium associated with the portfolio constructed using $ILLIQ^{TA}_{i,m}$ can be explained by that constructed using $ILLIQ^{A}_{i,m}$. The empirical results in the second row of Panel A indicate that the regression coefficients of α_{TA} are all significant and positive across both equal-weighted and value-weighted approaches. For example, for the value-weighted approach, the intercept coefficients for α_{TA} are all significant and positive across the three-factor (0.0006), four-factor (0.0008), five-factor (0.0006), and six-factor (0.0008) regression models. Overall, the results in Panel A of Table 7 suggest that the long–short strategy based on $ILLIQ^{A}_{i,m}$ encompasses the profitability of the long–short strategy based on $ILLIQ^{A}_{i,m}$. By contrast, the $ILLIQ^{A}_{i,m}$-based long–short strategy does not subsume the profitability of the $ILLIQ^{TA}_{i,m}$-based

long–short strategy.

Panel B of Table 7 presents the results pertaining to the relative profitability of long–short trading strategies based on Amihud's illiquidity ratio ($ILLIQ^{A}_{i,m}$) and daytime Amihud's illiquidity ratio ($ILLIQ^{DA}_{i,m}$), aiming to identify the ratio that more accurately captures the illiquidity premium in the Taiwan stock market. For the results in the first row of Panel B of Table 7, we set $\zeta = A$ and $\xi = DA$ to explore whether the long–short profitability constructed using $ILLIQ^{A}_{i,m}$ can be explained by that constructed using $ILLIQ^{DA}_{i,m}$. The empirical results suggest that the regression coefficients of the intercept are not statistically significant in both the equal-weighted and market-value-weighted approaches. For example, in the value-weighted analysis, the regression coefficients for the three-factor, four-factor, five-factor, and six-factor models are −0.0006, 0.0004, 0.0000, and 0.0011, respectively, with none being statistically significant. For the results in the second row of Panel B in Table 7, we set $\zeta = DA$ and $\xi = A$ to investigate whether the profitability of the long–short strategy constructed using $ILLIQ^{DA}_{i,m}$ can be explained by the profitability strategy constructed using $ILLIQ^{A}_{i,m}$. The findings reveal that, with the exception of the six-factor model, the intercept regression coefficients from the three-factor, four-factor, and five-factor models are statistically significant and positive in both the equal-weighted and market-value-weighted analyses. For example, in the value-weighted analysis, the intercept regression coefficients for the three-factor (0.0021), four-factor (0.0016), and five-factor (0.0017) models are significant. Overall, the results in Panel B of Table 7 indicate that the long-short profitability constructed using $ILLIQ^{DA}_{i,m}$ can subsume the profitability constructed using $ILLIQ^{A}_{i,m}$, but not vice versa.

Based on the findings obtained using $ILLIQ^{TA}_{i,m}$ and $ILLIQ^{DA}_{i,m}$, this paper argues that excluding overnight returns while incorporating the recency effect for the illiquidity ratio provides a more accurate measure than the original Amihud's illiquidity ratio for capturing the illiquidity premium. Panel C of Table 7 presents the results pertaining to the relative profitability of long–short trading strategies employing Amihud's illiquidity ratio ($ILLIQ^{A}_{i,m}$) and the time-weighted daytime Amihud's illiquidity ratio ($ILLIQ^{DA}_{i,m}$), with the purpose of identifying which metric provides a more accurate representation of the illiquidity premium in the Taiwan stock market. For the results in the first row of Panel C in Table 7, we set the dependent variable y as $R^{A}_{LS,K,m}$ and the independent variable x as $R^{TDA}_{LS,K,m}$ (i.e., let $\zeta = A$ and $\xi = TDA$) while controlling for other risk factors. This setup enables us to explore whether the profitability of the long–short strategy constructed using $ILLIQ^{A}_{i,m}$ can be explained by the profitability of the strategy based on $ILLIQ^{TDA}_{i,m}$. That is, we examine whether $R^{TDA}_{LS,K,m}$ can explain $R^{A}_{LS,K,m}$ while controlling for other risk factors. The empirical results indicate that, regardless of whether the equal- or value-weighted approaches are used, none of the intercepts is significantly positive. For the results in the second row of Panel C in Table 7, we reverse the settings for ζ and ξ (i.e., $\zeta = TDA$ and $\xi = A$) to verify whether the profitability of the long–short strategy constructed using $ILLIQ^{TDA}_{i,m}$ can be explained by the profitability of the strategy based on $ILLIQ^{A}_{i,m}$. Regardless of whether the equal-weighted method or value-weighted method is used, all intercept coefficients are significantly positive. For example, in the equal-weighted results, the regression coefficients of the intercept for the three-factor (0.0032), four-factor (0.0026), five-factor (0.0031), and six-factor (0.0024) models are all significant. These empirical results indicate that the illiquidity premium constructed using $ILLIQ^{TDA}_{i,m}$ subsumes the illiquidity premium constructed using $ILLIQ^{A}_{i,m}$, but not vice versa.

The study further explores whether the illiquidity measure that simultaneously excludes overnight returns and incorporates the recency effect encompasses measures that either consider the recency effect alone or exclude overnight returns alone. The empirical results are presented in Panels D and E of Table 7. Panel D presents the results pertaining to the profitability of long–short strategies, comparing those obtained using $ILLIQ^{TA}_{i,m}$ and $ILLIQ^{TDA}_{i,m}$. For the results in the first row of Panel D, we set $\zeta = TA$ and $\xi = TDA$ to investigate whether $R^{TA}_{LS,K,m}$ can be explained by $R^{TDA}_{LS,K,m}$ while controlling for other risk factors. Regardless of whether equal-weighted or value-weighted approaches are used, the empirical findings suggest that the intercepts of the regression coefficients for the three-factor, four-factor, five-factor, and six-factor models are insignificant. For the results in the second row of Panel D, this paper sets $\zeta = TDA$ and $\xi = TA$ to determine whether $R^{TDA}_{LS,K,m}$ can be explained by $R^{TA}_{LS,K,m}$ while controlling for other risk factors. Regardless of whether equal-weighted or value-weighted approaches are used, the results indicate that the regression coefficients of the intercepts are all significantly positive at the conventional level for the three-factor, four-factor, five-factor, and six-factor models. For example, when the equal-weighted approach is used, the regression coefficients of the intercepts for the three-factor, four-factor, five-factor, and six-factor models are 0.0026, 0.0018, 0.0025, and 0.0016, respectively, and all these coefficients are statistically significant. In summary, the empirical evidence presented in Panel D of Table 7 indicates that the $ILLIQ^{TDA}_{i,m}$-based long–short strategy subsumes the $ILLIQ^{TA}_{i,m}$-based long–short strategy, but not the other way around.

Panel E of Table 7 presents the results pertaining to the profitability of long–short strategies, comparing those obtained using $ILLIQ^{DA}_{i,m}$ and $ILLIQ^{TDA}_{i,m}$. For the results in the first row of Panel E, this paper sets $\zeta = DA$ and $\xi = TDA$ to investigate whether $R^{DA}_{LS,K,m}$ can be explained by $R^{TDA}_{LS,K,m}$ while controlling for other risk factors. Regardless of whether equal-weighted or value-weighted approaches are used, the empirical findings suggest that the intercepts of the regression coefficients for the three-factor, four-factor, five-factor, and six-factor models are significantly negative. For example, when the value-weighted approach is used, the intercept regression coefficients for the three-factor, four-factor, five-factor, and six-factor models are −0.0007, −0.0006, −0.0010, and − 0.0008, respectively, and all coefficients are statistically significant. For the results in the second row of Panel E, we set $\zeta = TDA$ and $\xi = DA$ to investigate whether $R^{TDA}_{LS,K,m}$ can be explained by $R^{DA}_{LS,K,m}$ while controlling for other risk factors. The results indicate that the regression coefficients for the intercept are all significantly positive for the three-factor, four-factor, five-factor, and six-factor models, regardless of the weighting approach used. Overall, the results indicate that the profitability achieved through the $ILLIQ^{TDA}_{i,m}$-based long–short strategy subsumes the profitability of the $ILLIQ^{DA}_{i,m}$-based long–short strategy, but not vice versa.

3.3. VAR analysis

Subsection 3.2, which discusses asset pricing tests, reveals that the optimal capture of illiquidity premiums in the Taiwan stock market can be achieved by considering time weighting and removing overnight returns. In this subsection, we implement the methodology of Amihud (2002) to investigate whether the proposed illiquidity measure, which incorporates time weighting and excludes overnight returns, exhibits improved predictive power for the overall market compared with the traditional Amihud's illiquidity measure. The following VAR model is used:

$$R_m^{MKT} = \alpha_1 + \sum_{l=1}^{L} \beta_{1l} R_{m-l}^{MKT} + \sum_{l=1}^{L} \delta_{1l} AILLIQ_{m-l}^{A} + \sum_{l=1}^{L} \gamma_{1l} AILLIQ_{m-l}^{TDA} + \epsilon_m^{MKT} \tag{14.1}$$

$$AILLIQ_m^{A} = \alpha_2 + \sum_{l=1}^{L} \beta_{2l} R_{m-l}^{MKT} + \sum_{l=1}^{L} \delta_{2l} AILLIQ_{m-l}^{A} + \sum_{l=1}^{L} \gamma_{2l} AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^A} \tag{14.2}$$

$$AILLIQ_m^{TDA} = \alpha_3 + \sum_{l=1}^{L} \beta_{3l} R_{m-l}^{MKT} + \sum_{l=1}^{L} \delta_{3l} AILLIQ_{m-l}^{A} + \sum_{l=1}^{L} \gamma_{3l} AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^{TDA}} \tag{14.3}$$

where R_m^{MKT} is the market return in month m. $AILLIQ_m^{A}$ is the average of Amihud's illiquidity ratio across stocks in each month and is calculated as

$$AILLIQ_m^{A} = \frac{1}{N_m} \sum_{i=1}^{N_m} ILLIQ_{i,m}^{A} \tag{15}$$

where N_m is the number of stocks in month m. $ILLIQ_{i,m}^{A}$ is Amihud's illiquidity ratio for stock i in month m. $AILLIQ_m^{TDA}$ is the average of the time-weighted daytime Amihud's illiquidity ratio in each month and is calculated in an analogous manner to $AILLIQ_m^{A}$ as:

$$AILLIQ_m^{TDA} = \frac{1}{N_m} \sum_{i=1}^{N_m} ILLIQ_{i,m}^{TDA} \tag{16}$$

where N_m is the number of stocks in month m and $ILLIQ_{i,m}^{TDA}$ is the time-weighted daytime Amihud's illiquidity ratio for stock i in month m. ϵ_m^{MKT}, $\epsilon_m^{ILLIQ^A}$, and $\epsilon_m^{ILLIQ^{TDA}}$ are the residuals for R_m^{MKT}, $AILLIQ_m^{A}$, and $AILLIQ_m^{TDA}$. The VAR models enable us to examine the dynamic behavior among market returns and illiquidity measures by conducing Granger-causality tests and sum-of-coefficients tests. To ensure that the results are robust to the chosen number of lags, the study reports the VAR results for up to 12 lags, thereby enabling the causality and predictability for short- or long-time horizons to be distinguished (Dufour and Renault, 1998; Lee et al., 2010).

The VAR results are reported in Table 8. Panels A and B of Table 8 report results from the Granger-causality tests and sum-of-coefficients tests. These results indicate that both $AILLIQ_m^{A}$ and $AILLIQ_m^{TDA}$ Granger-cause R_m^{MKT} for most lags. Specifically, most of the Granger-causality tests for $\delta_{1l}(L) = 0$ and $\gamma_{1l}(L) = 0$ are significant at the 1% level. For instance, the $F-$ statistics of $\delta_{1l}(L) = 0$ are 5.4527, 5.4666, and 6.9744 for 1-, 6-, and 12-lag, respectively. However, R_m^{MKT} does not Granger-cause $AILLIQ_m^{A}$ or $AILLIQ_m^{TDA}$ because the Granger-causality test results for $\beta_{2l}(L) = 0$ and $\beta_{3l}(L) = 0$ across different lags are insignificant at the conventional level. Additionally, the results indicate two-way Granger causality between $AILLIQ_m^{A}$ and $AILLIQ_m^{TDA}$ for most lags. Panel B of Table 8 presents the sum-of-coefficients test results. The empirical results provide strong evidence of the significant and positive effect of $AILLIQ_m^{TDA}$ on subsequent R_m^{MKT} for most lags (6-, 9-, and 12-lag). However, the sum-of-coefficients tests reveal the effect of $AILLIQ_m^{A}$ on subsequent R_m^{MKT} is only significantly positive over the long horizon (e.g., a lag of 12 months).

Because aggregate market returns are not practically tradable, the study employs returns from stock index futures to conduct VAR analysis. The VAR model is estimated as follows:

$$R_m^{FUT} = \alpha_1 + \sum_{l=1}^{L} \beta_{1l} R_{m-l}^{FUT} + \sum_{l=1}^{L} \delta_{1l} AILLIQ_{m-l}^{A} + \sum_{l=1}^{L} \gamma_{1l} AILLIQ_{m-l}^{TDA} + \epsilon_m^{FUT} \tag{17.1}$$

$$AILLIQ_m^{A} = \alpha_2 + \sum_{l=1}^{L} \beta_{2l} R_{m-l}^{FUT} + \sum_{l=1}^{L} \delta_{2l} AILLIQ_{m-l}^{A} + \sum_{l=1}^{L} \gamma_{2l} AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^A} \tag{17.2}$$

$$AILLIQ_m^{TDA} = \alpha_3 + \sum_{l=1}^{L} \beta_{3l} R_{m-l}^{FUT} + \sum_{l=1}^{L} \delta_{3l} AILLIQ_{m-l}^{A} + \sum_{l=1}^{L} \gamma_{3l} AILLIQ_{m-l}^{TDA} + \epsilon_m^{ILLIQ^{TDA}} \tag{17.3}$$

where R_m^{FUT} is the futures return in month m. ϵ_m^{FUT} is the residual of R_m^{FUT}. This paper utilizes the futures index associated with the Taiwan Weighted Stock Index, which is traded on the Taiwan Futures Exchange (TAIFEX), for analysis. The results are reported in Table 9.

Panel A of Table 9 presents the results of the Granger-causality tests. The evidence indicates that even though most of the Granger-

causality tests for $\delta_{1l}(L) = 0$ and $\gamma_{1l}(L) = 0$ yield significant results at the 1% level, the Granger-causality tests for $\beta_{2l}(L) = 0$ and $\beta_{3l}(L) = 0$ across different lags are insignificant at the conventional level. These findings suggest that the illiquidity measure, irrespective of whether $AILLIQ_m^A$ or $AILLIQ_m^{TDA}$ is used, Granger-causes R_m^{FUT}, but not vice versa. The empirical results also indicate the presence of a bi-directional Granger-causality relationship between $AILLIQ_m^A$ and $AILLIQ_m^{TDA}$ for most lags. Panel B of Table 9 reports the sum-of-coefficients test results. These empirical results further provide strong evidence of the positive effect of $AILLIQ_m^{TDA}$ on subsequent index futures returns for most lags (6-, 9-, and 12-lag), and this effect is more pronounced over the long-horizon. By contrast, the positive effect of $AILLIQ_m^A$ on subsequent index futures returns is significantly positive but primarily observed over a long horizon. Overall, Tables 8 and 9 indicate that compared with $AILLIQ_m^A$, $AILLIQ_m^{TDA}$ has a greater effect on subsequent market returns, such as R_m^{MKT} and R_m^{FUT}, and its positive effect can persist over short- to long-time horizons.

3.4. Robustness analysis

As revealed in Tables 6 to 9, the time-weighted daytime Amihud's illiquidity ratio proposed in the present study has a more pronounced influence on subsequent stock returns under both cross-sectional and time-series approaches compared to other illiquidity measures. This finding is further supported by the VAR analysis results. As a robustness check, the study employs $\alpha = 1$ to calculate the weighting function, further confirming our analysis. The results are displayed in Tables 10 to 12.

Table 10 presents the robustness check for the Fama–Macbeth (Fama and MacBeth, 1973) mean monthly coefficients for the effects of the four illiquidity measures on subsequent stock returns while controlling *SIZE*, *BM*, and *MOM*. Models (1) to (4) examine the cross-sectional effects of $ILLIQ_{i,m}^A$, $ILLIQ_{i,m}^{TA}$, $ILLIQ_{i,m}^{DA}$, and $ILLIQ_{i,m}^{TDA}$ on subsequent stock returns. The empirical results indicate that the regression coefficients for $ILLIQ_{i,m}^A$, $ILLIQ_{i,m}^{TA}$, $ILLIQ_{i,m}^{DA}$, and $ILLIQ_{i,m}^{TDA}$ are all significantly positive. Models (5), (6), and (7) assess whether the effect of $ILLIQ_{i,m}^A$ on subsequent stock returns remains significantly positive after including $ILLIQ_{i,m}^{TA}$, $ILLIQ_{i,m}^{DA}$, and $ILLIQ_{i,m}^{TDA}$. The results of Models (5), (6), and (7) indicate that after $ILLIQ_{i,m}^{TA}$, $ILLIQ_{i,m}^{DA}$, and $ILLIQ_{i,m}^{TDA}$ are incorporated separately, the coefficient for $ILLIQ_{i,m}^A$ becomes insignificantly positive, whereas the coefficients for the three liquidity measures ($ILLIQ_{i,m}^{TA}$, $ILLIQ_{i,m}^{DA}$, and $ILLIQ_{i,m}^{TDA}$) remain significantly positive. Models (8) and (9) investigate whether the effects of $ILLIQ_{i,m}^{TA}$ and $ILLIQ_{i,m}^{DA}$ on subsequent stock returns remain significantly positive after including $ILLIQ_{i,m}^{TDA}$. The empirical results indicate that after the inclusion of $ILLIQ_{i,m}^{TDA}$, the regression coefficients for $ILLIQ_{i,m}^{TA}$ and $ILLIQ_{i,m}^{DA}$ are no longer significantly positive, with the effect of $ILLIQ_{i,m}^{DA}$ changing from a significant positive to a significant negative coefficient. However, the regression coefficients of $ILLIQ_{i,m}^{TDA}$ for Models (8) and (9) remain significantly positive.

Table 11 reports the robustness check results for equal- and value-weighted long–short portfolio returns across various variables in time-series regressions. Panel A compares Amihud's illiquidity ratio with the time-weighted Amihud's illiquidity ratio. Panel B investigates Amihud's illiquidity ratio with the daytime Amihud's illiquidity ratio. Panel C examines Amihud's illiquidity ratio against the time-weighted daytime Amihud's illiquidity ratio. Panel D compares the time-weighted Amihud's illiquidity ratio with time-weighted daytime Amihud's illiquidity ratio. Panel E evaluates the daytime Amihud's illiquidity ratio versus time-weighted daytime Amihud's illiquidity ratio. In general, the conclusions drawn from the empirical results in Table 11 are completely consistent with those presented in Table 7. Specifically, regarding the profitability of the long–short strategy, the results in Panels A to C of Table 11 indicate that $ILLIQ_{i,m}^{TA}$ subsumes $ILLIQ_{i,m}^A$, and $ILLIQ_{i,m}^{DA}$ subsumes $ILLIQ_{i,m}^A$, which ultimately leads to $ILLIQ_{i,m}^{TDA}$ subsuming $ILLIQ_{i,m}^A$. Additionally, the empirical findings presented in Panels D and E of Table 11 indicate that $ILLIQ_{i,m}^{TDA}$ also encompasses $ILLIQ_{i,m}^{TA}$ (Panel D) and $ILLIQ_{i,m}^{DA}$ (Panel E).

Table 12 presents the VAR analysis results for futures returns and the illiquidity ratio measures. Panel A of Table 12 presents the results from the Granger-causality tests. The findings indicate that the Granger-causality tests for $\delta_{1l}(L) = 0$ and $\gamma_{1l}(L) = 0$ yield mostly significant results at the 1% level across various lags. Panel B of Table 12 reports the empirical results from the sum-of-coefficients tests, which strongly support the positive effect of $AILLIQ_m^{TDA}$ on subsequent index futures returns across both short and long horizons. By contrast, the positive effect of $AILLIQ_m^A$ on subsequent index futures returns is notably positive only over a long horizon. Overall, these findings support our conclusions, indicating that the time-weighted daytime Amihud's illiquidity ratio is more effective than the Amihud's illiquidity ratio in capturing the illiquidity premium in the Taiwan stock market.

4. Conclusions

The present study empirically analyzed stock market liquidity in the Taiwan stock market. Amihud (2002) illiquidity measure is widely adopted in asset pricing research. Amihud's illiquidity measure can be conveniently calculated by using close-to-close stock returns and dollar trading volume. Nevertheless, there are two reasons why Amihud (2002) illiquidity measure needs to be adjusted when used in the Taiwan stock market. The first concern relates to overnight returns. Barardehi et al. (2021) suggest that overnight returns are primarily influenced by information arrivals unrelated to the daily trading volume, which is used in the denominator. Consequently, Amihud (2002) illiquidity ratio incorporates overnight returns into a proxy for price impact, resulting in a fundamental time mismatch between the denominator and numerator. As noted by Fung et al. (2010), overnight returns in the Asian stock market tend to exhibit an overreaction to U.S. stock returns from the previous day. Given that the Taiwan stock market is heavily weighted towards technology stocks, its overnight returns are often swayed by movements in U.S. technology indices such as NASDAQ and

semiconductor sector indices. Thus, the issue of overnight returns is particularly relevant for the Taiwan stock market. The second issue with Amihud's illiquidity measure is that it uses equal-weighted daily ratios to calculate the illiquidity ratio over a given period. However, as indicated by Greenwood and Shleifer (2014), Barberis et al. (2018), and among others, recent observations (e.g., returns) have a more pronounced effect on investor expectations relative to observations from the more distant past because of the recency effect. Previous studies have indicated the presence of the recency effect in the Taiwan stock market. Hao et al. (2016) report evidence of the profitability of the recency-based strategies in Taiwan. Hence, the recency effect plays a crucial role in shaping investor behavior and market dynamics. These challenges associated with overnight returns and the recency effect significantly affect the calculation and interpretation of Amihud's illiquidity measure in the Taiwan stock market. Addressing these issues is essential for improving the precision and relevance of liquidity measurement.

To investigate the effects of overnight returns and the recency effect on the calculation of illiquidity ratios in the Taiwan stock market, this study uses four illiquidity measures for our analysis. First, we consider Amihud's illiquidity ratio based on close-to-close returns (Amihud, 2002) and the daytime Amihud illiquidity ratio on the basis of open-to-close returns (Barardehi et al., 2021). Subsequently, we propose adjusted ratios that account for the recency effect by using these two measures. As such, the four illiquidity measures are Amihud's illiquidity ratio, the time-weighted Amihud's illiquidity ratio, the daytime Amihud's illiquidity ratio, and the time-weighted daytime Amihud's illiquidity ratio. The study then assesses the effectiveness of these measures through asset pricing tests, conducting both cross-sectional and time-series analyses.

Using data from the Taiwan stock market for the period from January 2008 to October 2023, the study reveals that the profitability of long–short strategies is highest when using the time-weighted daytime Amihud's illiquidity measure, followed by the daytime Amihud's illiquidity measure, time-weighted Amihud's illiquidity measure, and Amihud's illiquidity measure. Furthermore, the paper discovers that overnight returns and the recency effect significantly influence the computation of illiquidity measures. Specifically, whether excluding overnight returns alone or accounting for the recency effect alone, both methods consistently offer more accurate representations of the illiquidity premium in the Taiwan stock market compared to Amihud's illiquidity ratio. Building on the aforementioned findings, this study further finds that the time-weighted daytime Amihud's illiquidity ratio can explain stock returns more effectively and represent illiquidity premiums more accurately compared with the traditional Amihud's illiquidity ratio and other illiquidity ratios that only consider the exclusion of overnight returns or account for the recency effect. These findings are consistent across both cross-sectional and time-series tests. Finally, this study employs VAR analysis to investigate the lead–lag relationship between aggregate market returns and illiquidity measures. The findings suggest that both Amihud's illiquidity measure and the time-weighted daytime Amihud's illiquidity measure Granger-cause aggregate market returns, with no reverse causality detected. Notably, the study discovers that the positive effect of the time-weighted daytime Amihud's illiquidity measure is significantly more pronounced than Amihud's traditional illiquidity measure. In summary, our empirical results indicate that both overnight returns and the recency effect have significant effects when calculating illiquidity measure. Consequently, this paper proposes a novel, theoretically grounded liquidity metric that provides valuable insights for academic researchers and practitioners not only in the Taiwan stock market but also in other Asian stock markets.

CRediT authorship contribution statement

Hsiu-Chuan Lee: Writing – original draft, Software, Methodology, Funding acquisition, Conceptualization. **Donald Lien:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Her-Jiun Sheu:** Writing – review & editing, Validation, Conceptualization. **Chung-Jen Yang:** Software, Methodology, Conceptualization.

Acknowledgments

The authors would like to thank an anonymous referee for helpful comments and suggestions that significantly improved this paper. Hsiu-Chuan Lee is grateful for the financial support from the National Science and Technology Council of Taiwan (NSTC 112-2410-H-130-031-).

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